

Related U.S. Application Data

- which is a continuation of application No. 18/241,633, filed on Sep. 1, 2023, now Pat. No. 11,918,869, which is a continuation-in-part of application No. 18/219,215, filed on Jul. 7, 2023, now Pat. No. 11,839,801.
- (60) Provisional application No. 63/708,955, filed on Oct. 18, 2024, provisional application No. 63/672,521, filed on Jul. 17, 2024, provisional application No. 63/525,847, filed on Jul. 10, 2023, provisional application No. 63/524,452, filed on Jun. 30, 2023, provisional application No. 63/470,711, filed on Jun. 2, 2023.

(52) **U.S. Cl.**

CPC *A63B 53/0408* (2020.08); *A63B 53/0437* (2020.08); *A63B 53/0441* (2020.08); *A63B 53/0466* (2013.01); *A63B 53/047* (2013.01); *A63B 2053/0491* (2013.01)

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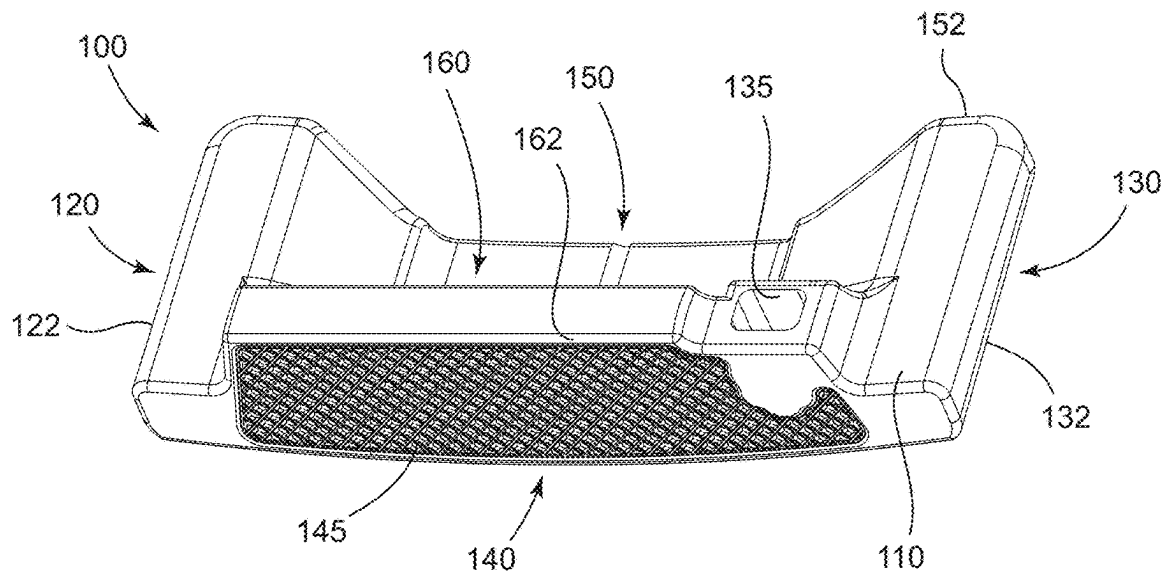


FIG. 1

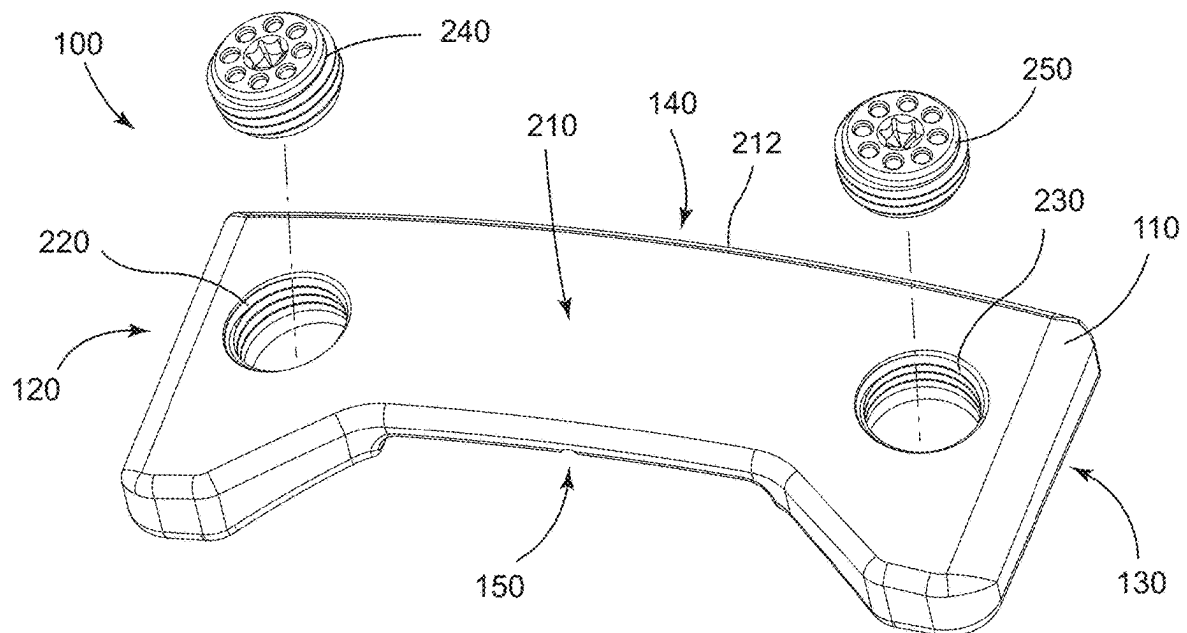


FIG. 2

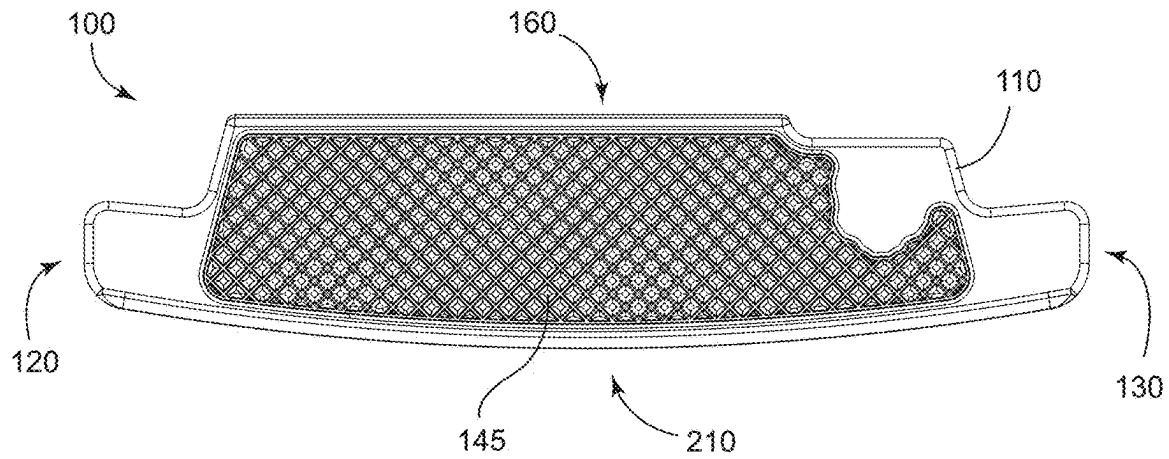


FIG. 3

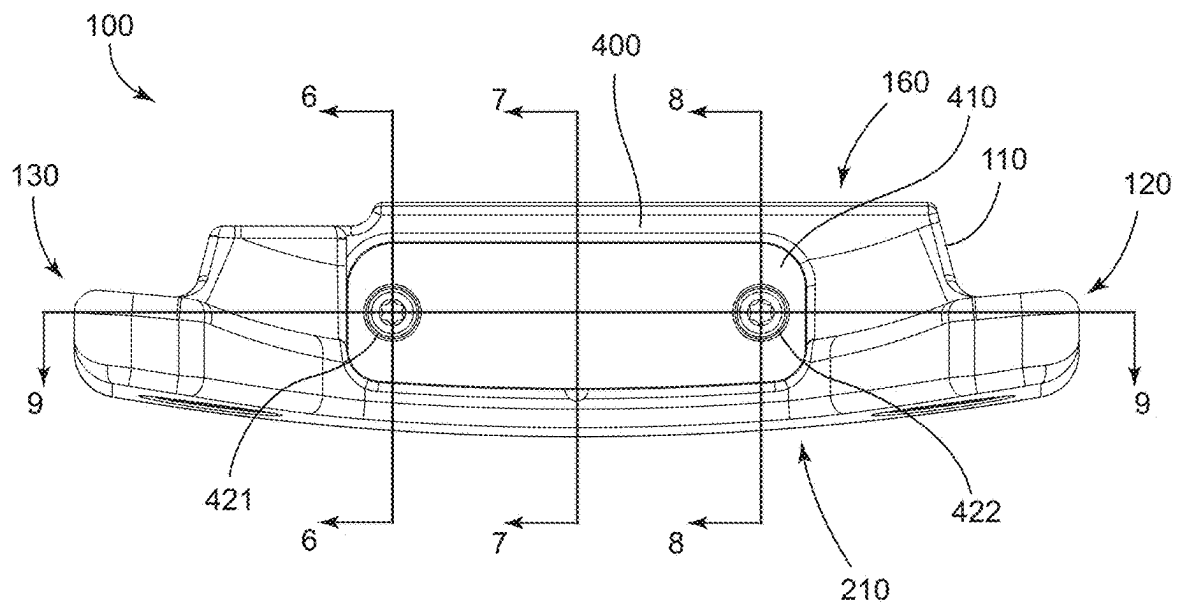


FIG. 4

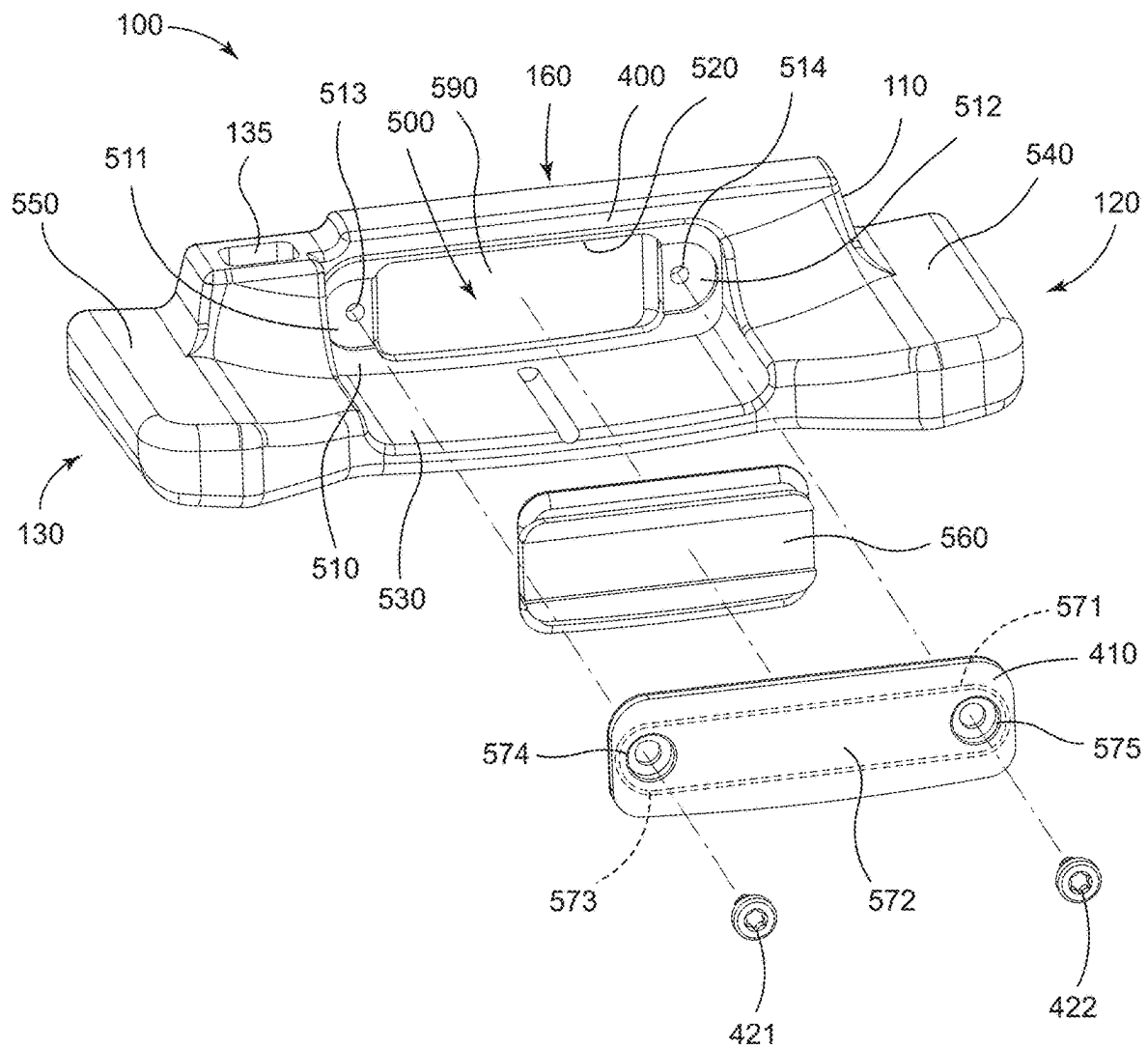


FIG. 5

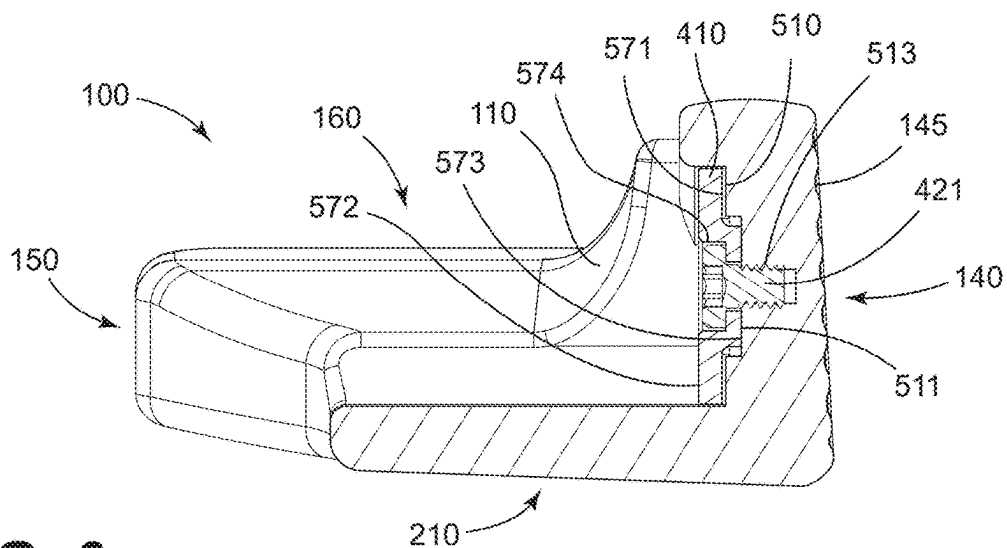


FIG. 6

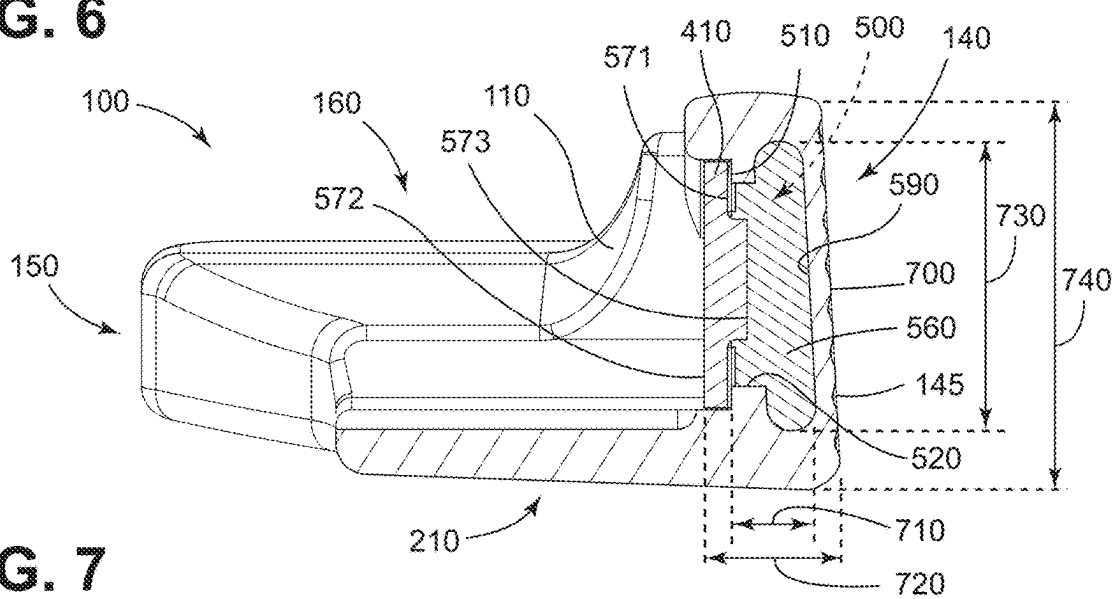


FIG. 7

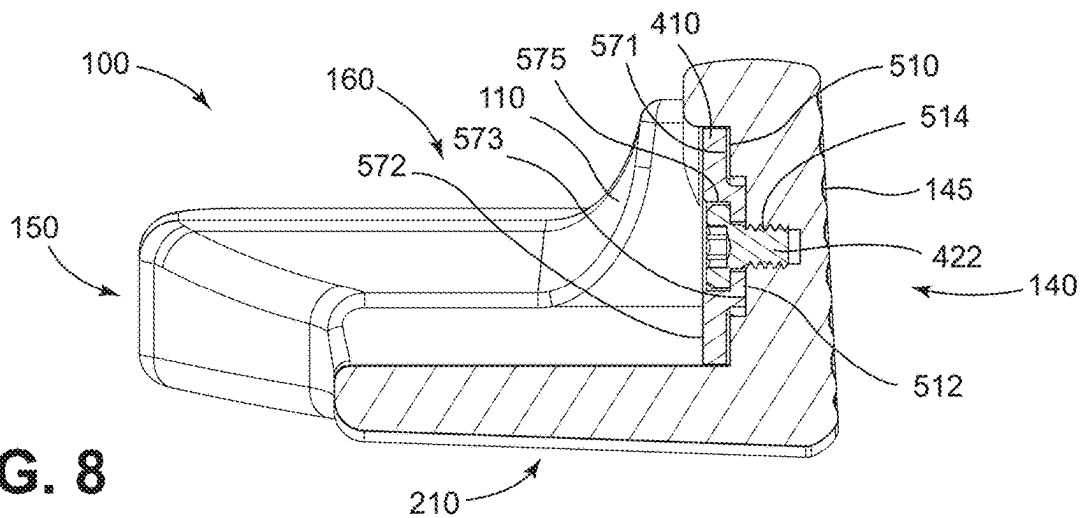


FIG. 8

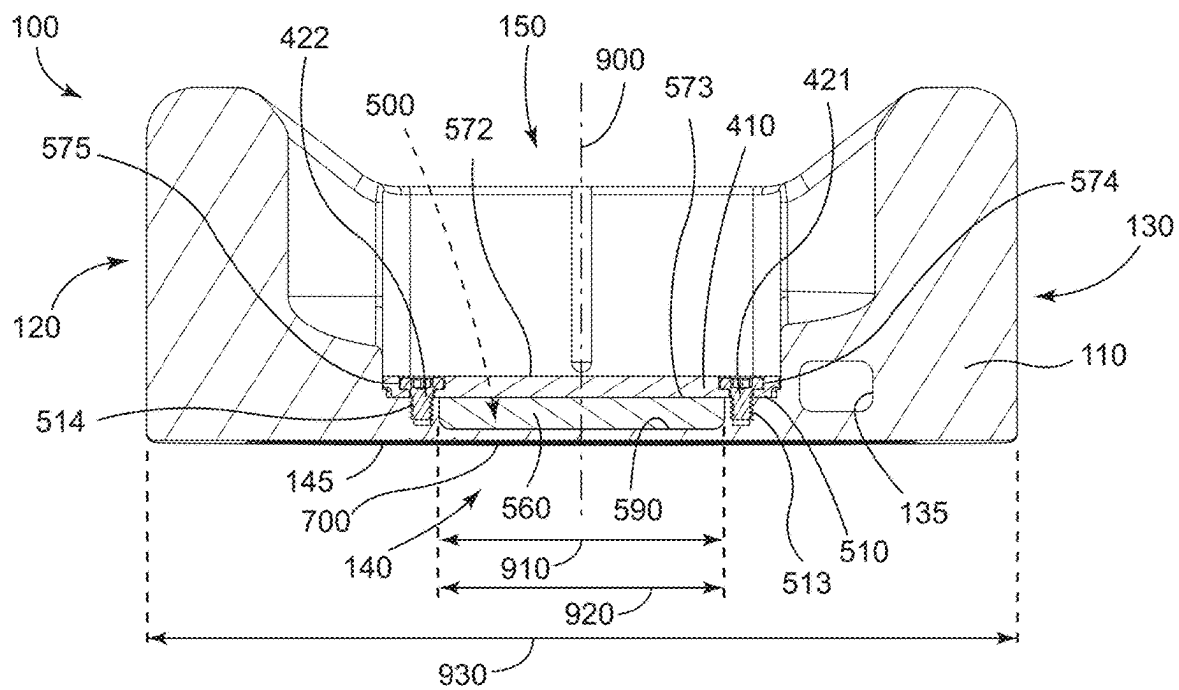


FIG. 9

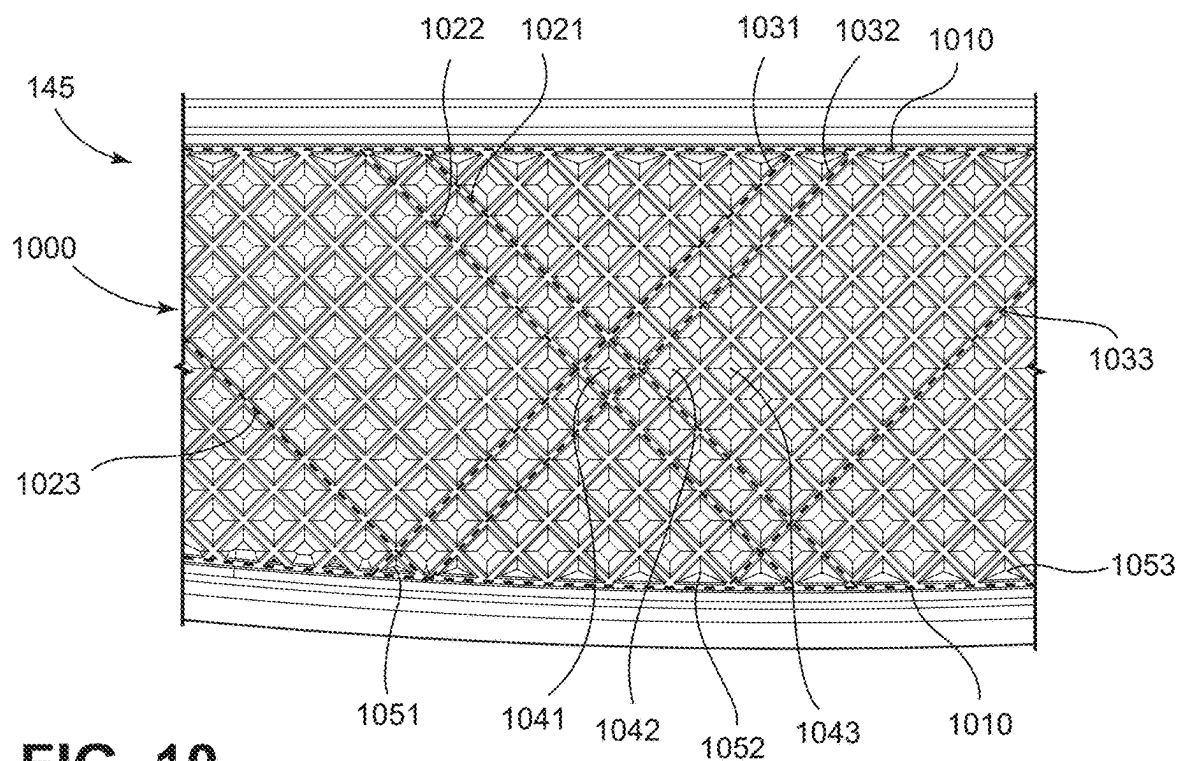


FIG. 10

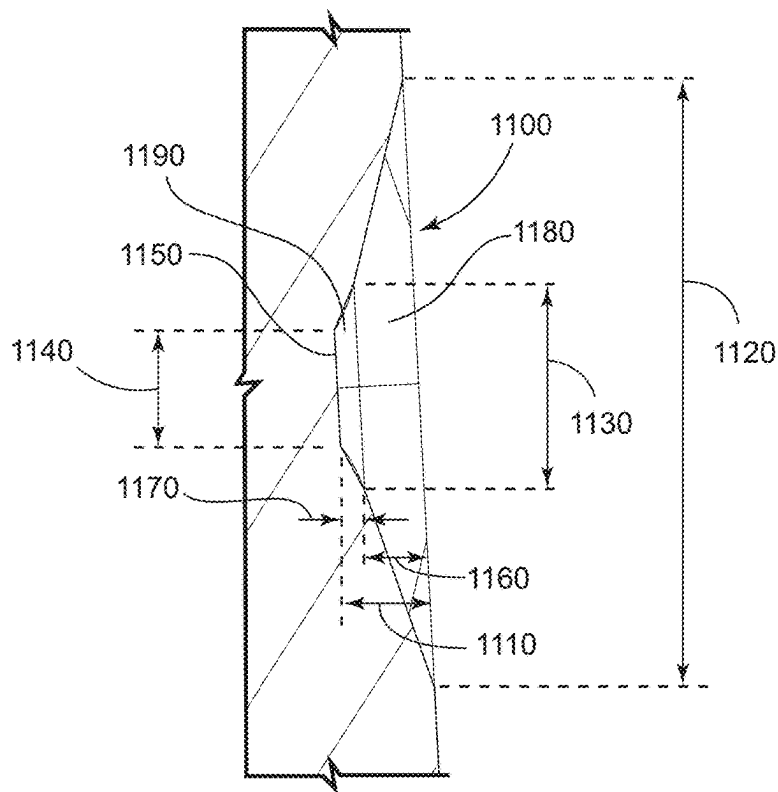


FIG. 11

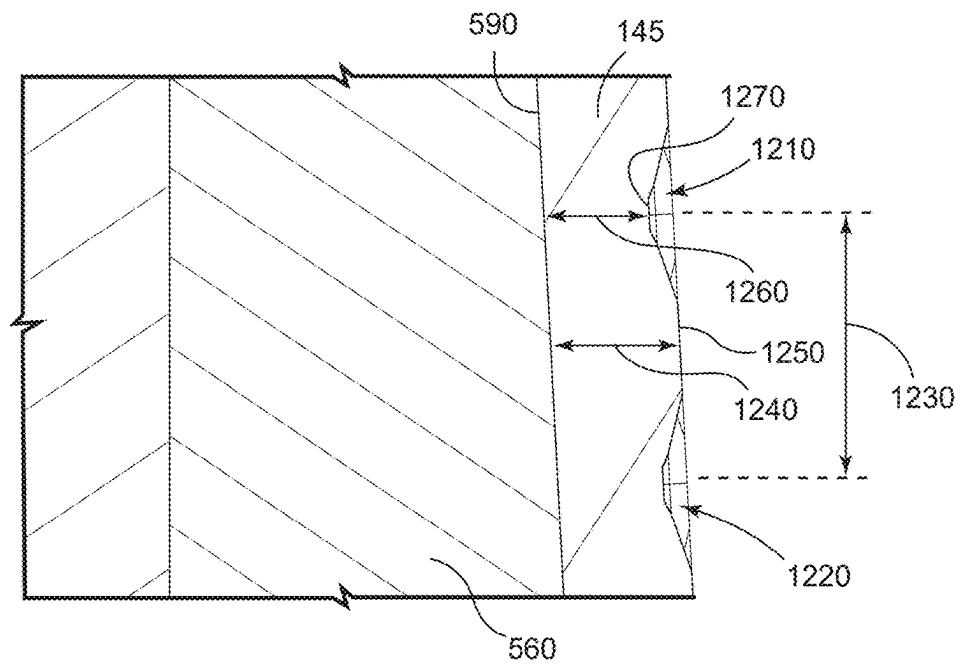
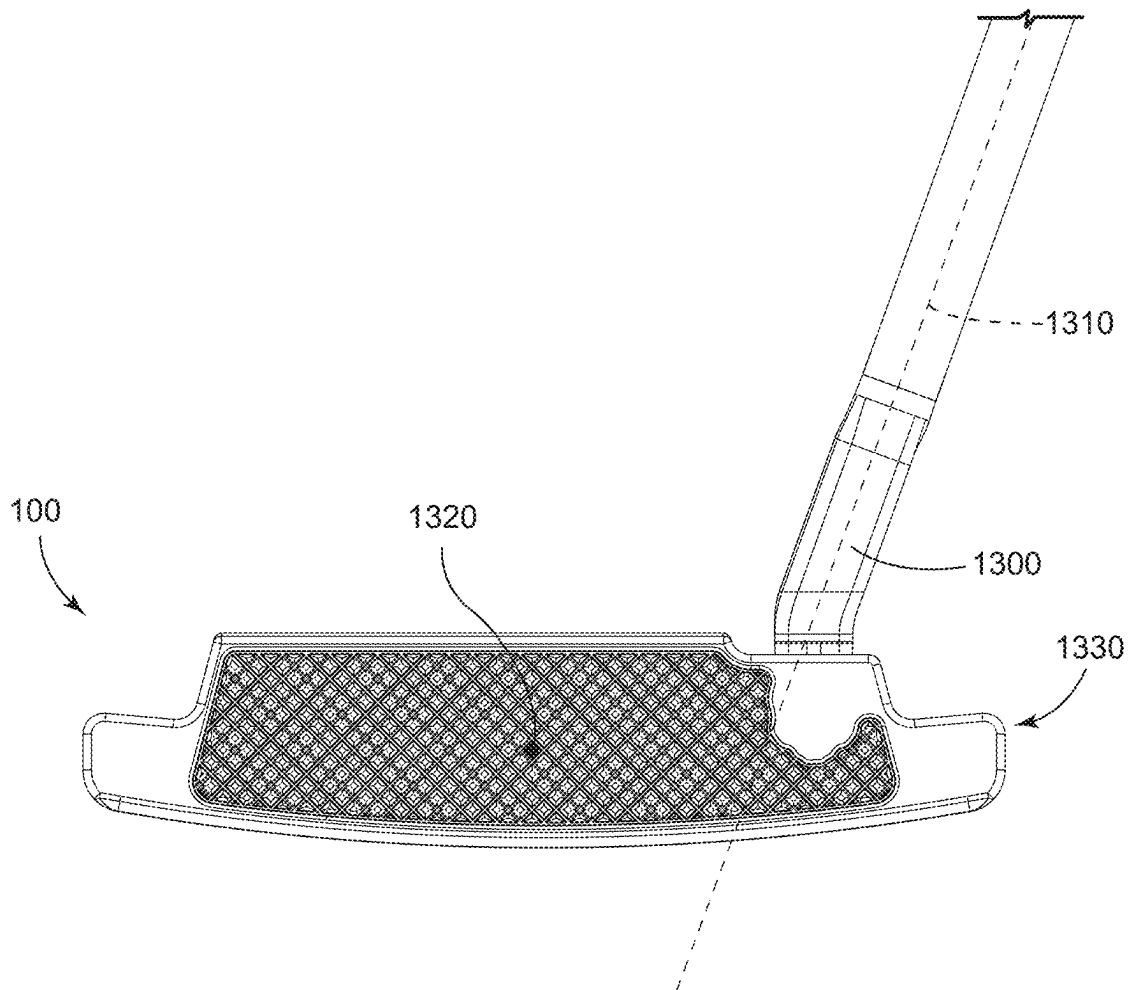
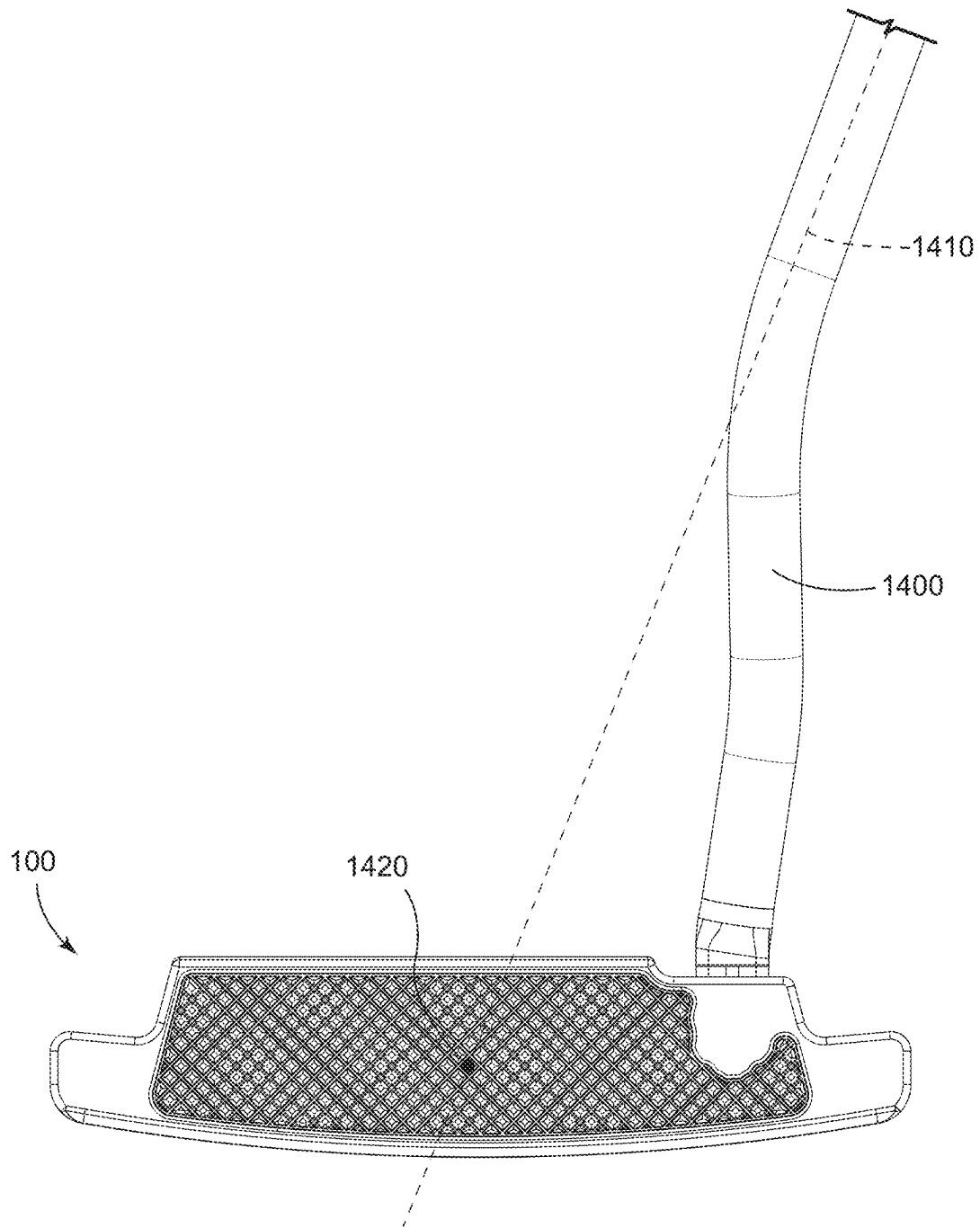
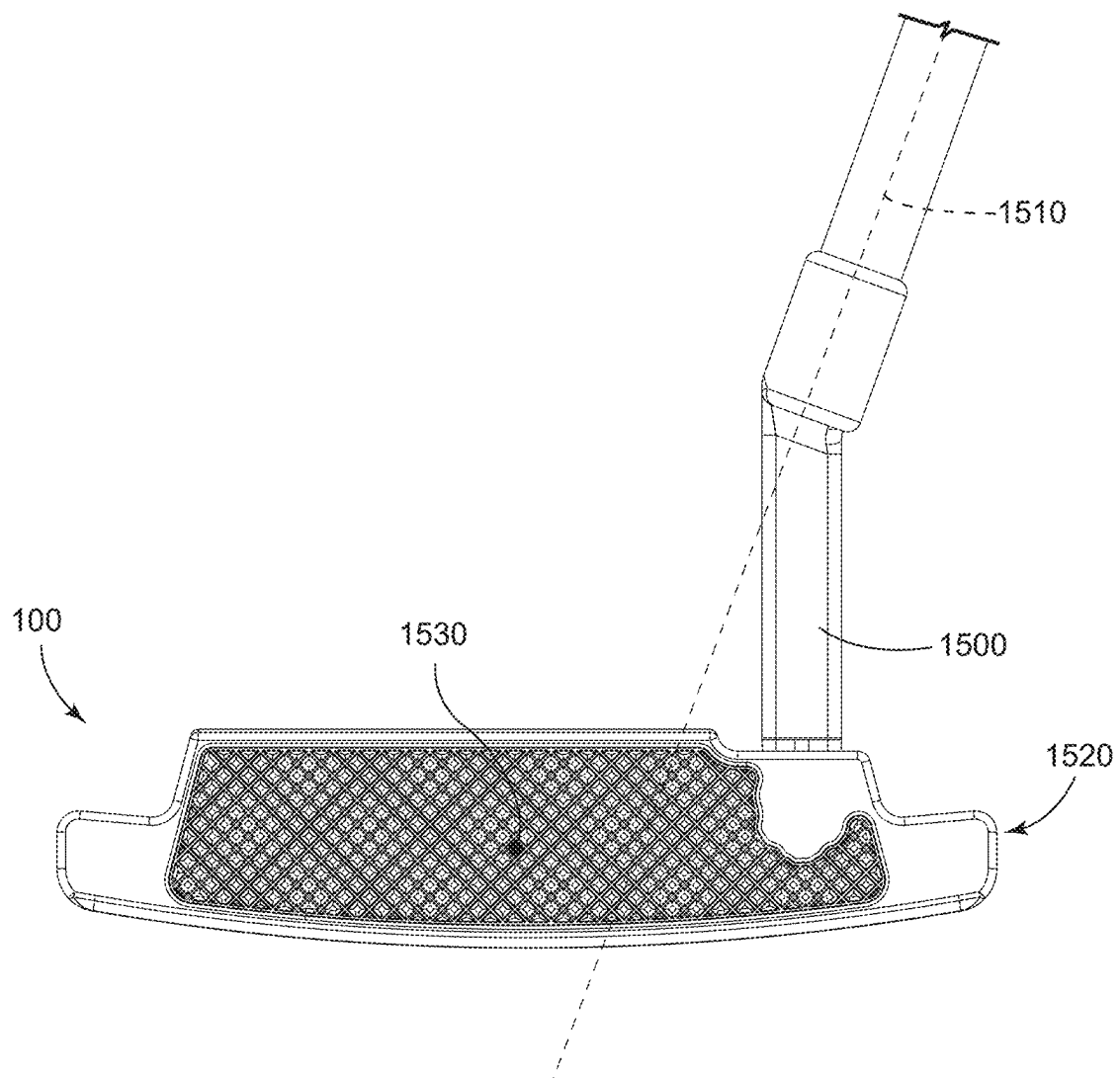


FIG. 12

**FIG. 13**

**FIG. 14**

**FIG. 15**

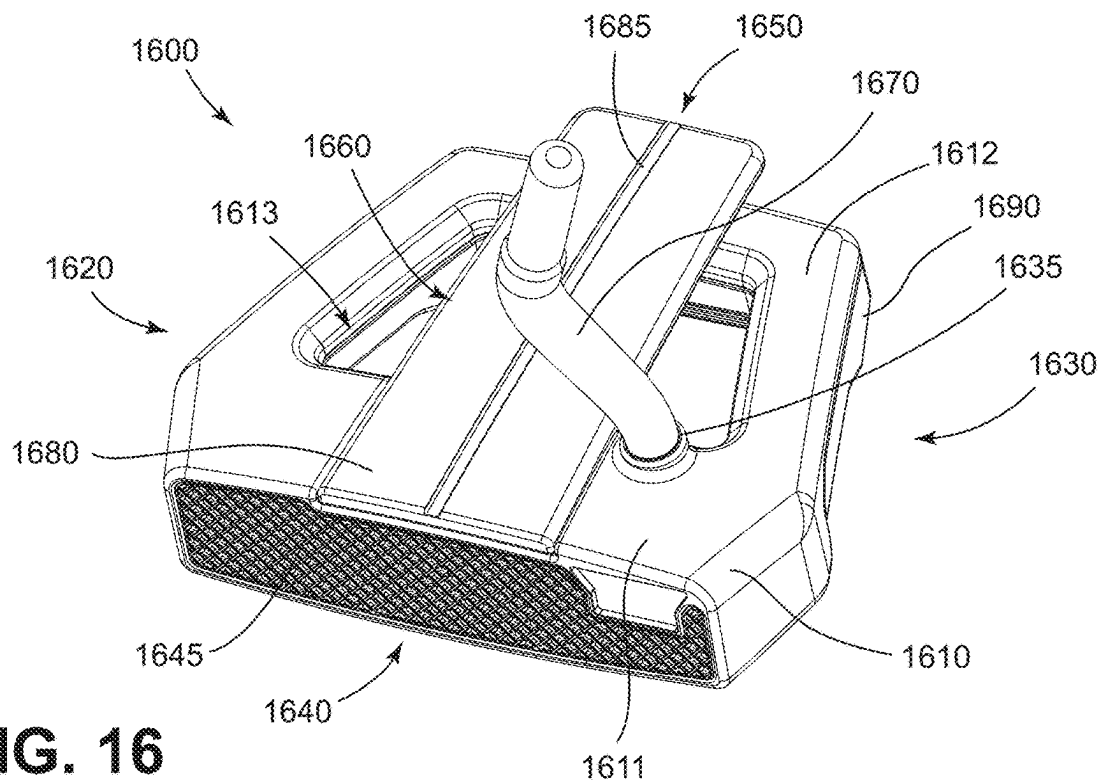


FIG. 16

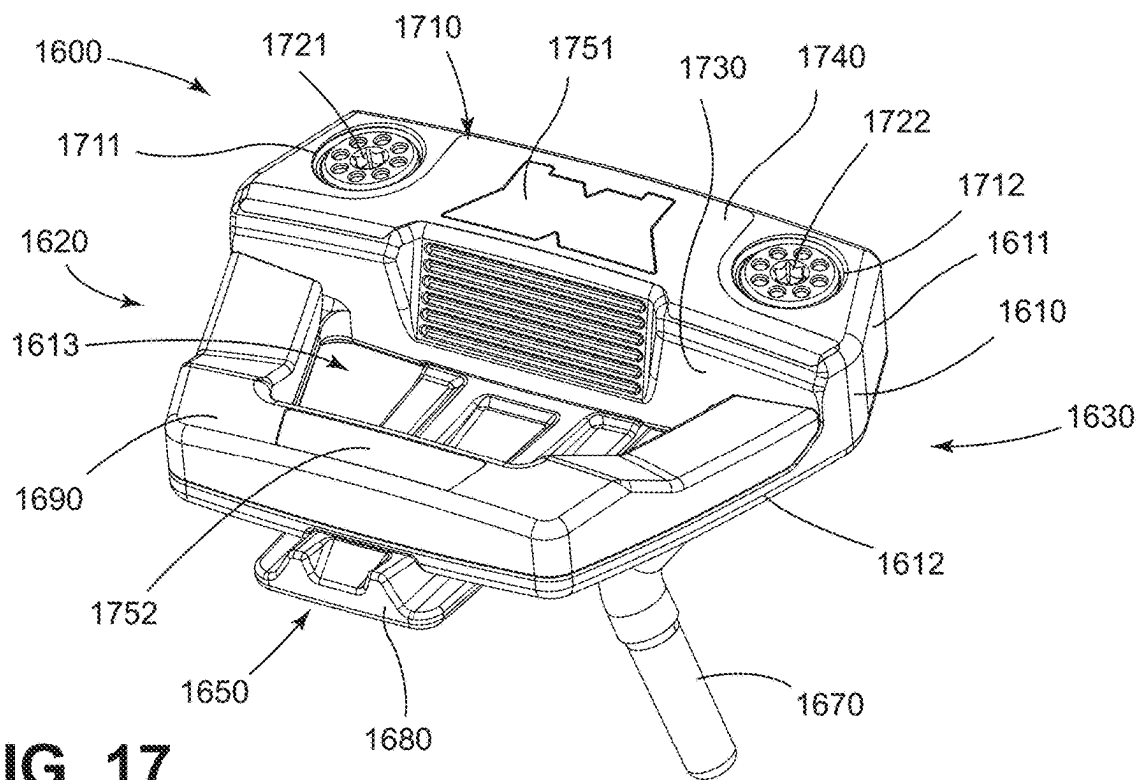


FIG. 17

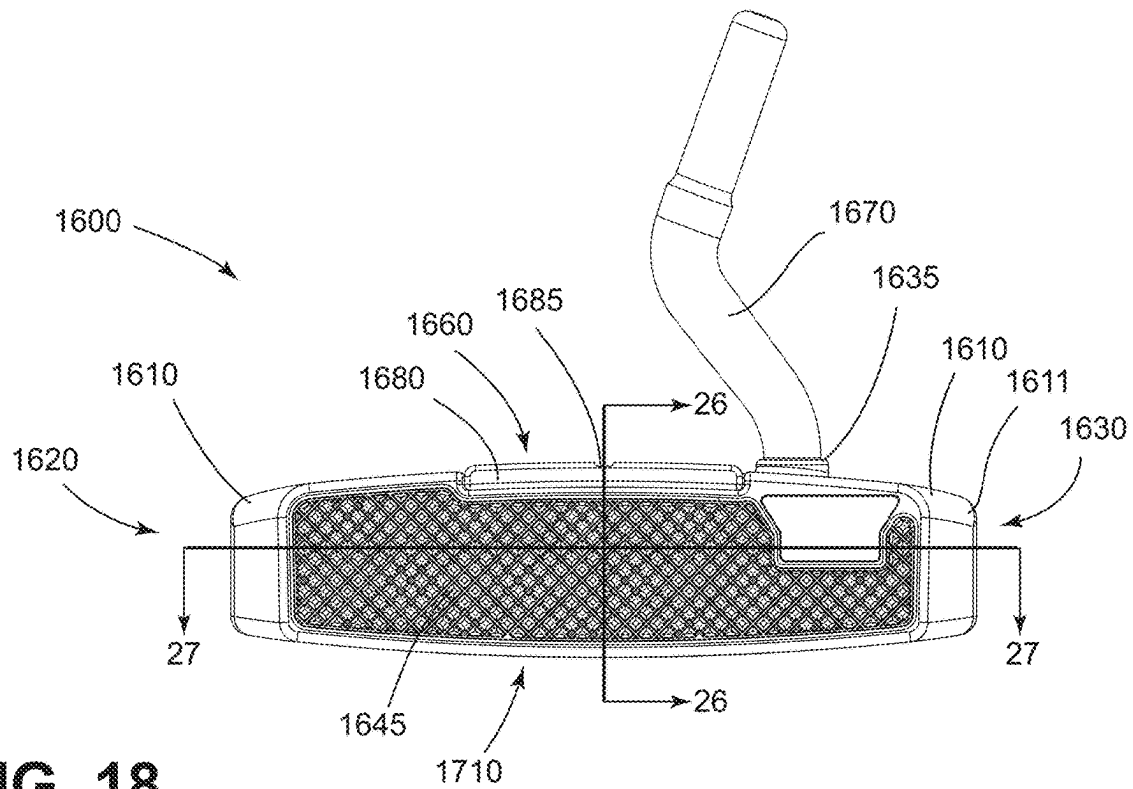


FIG. 18

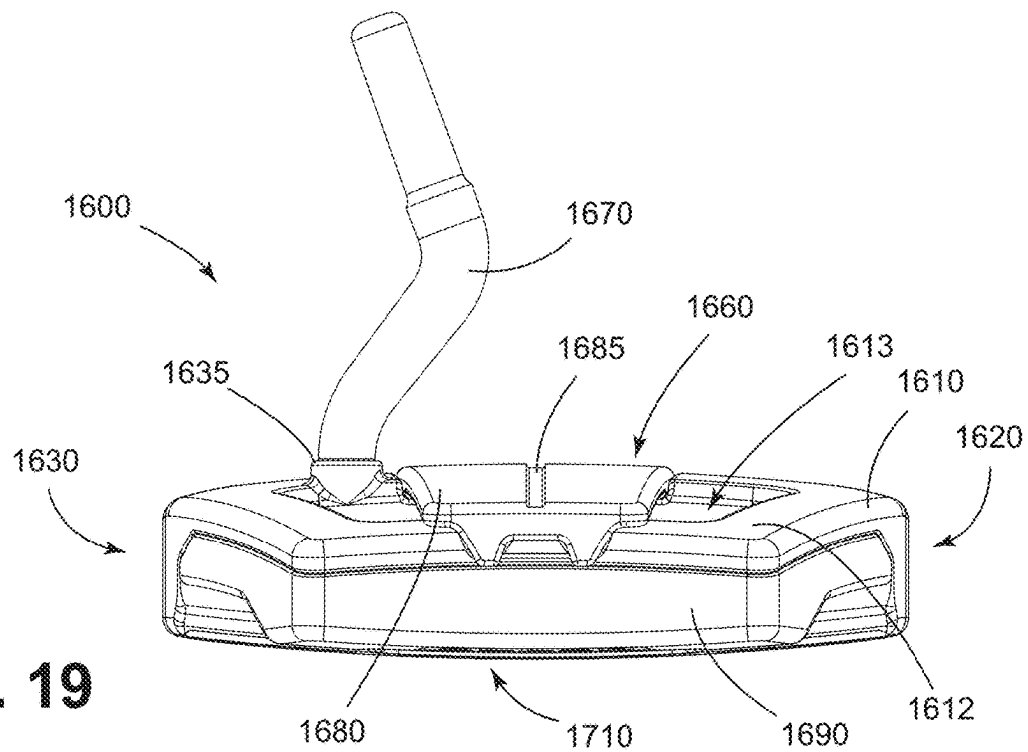
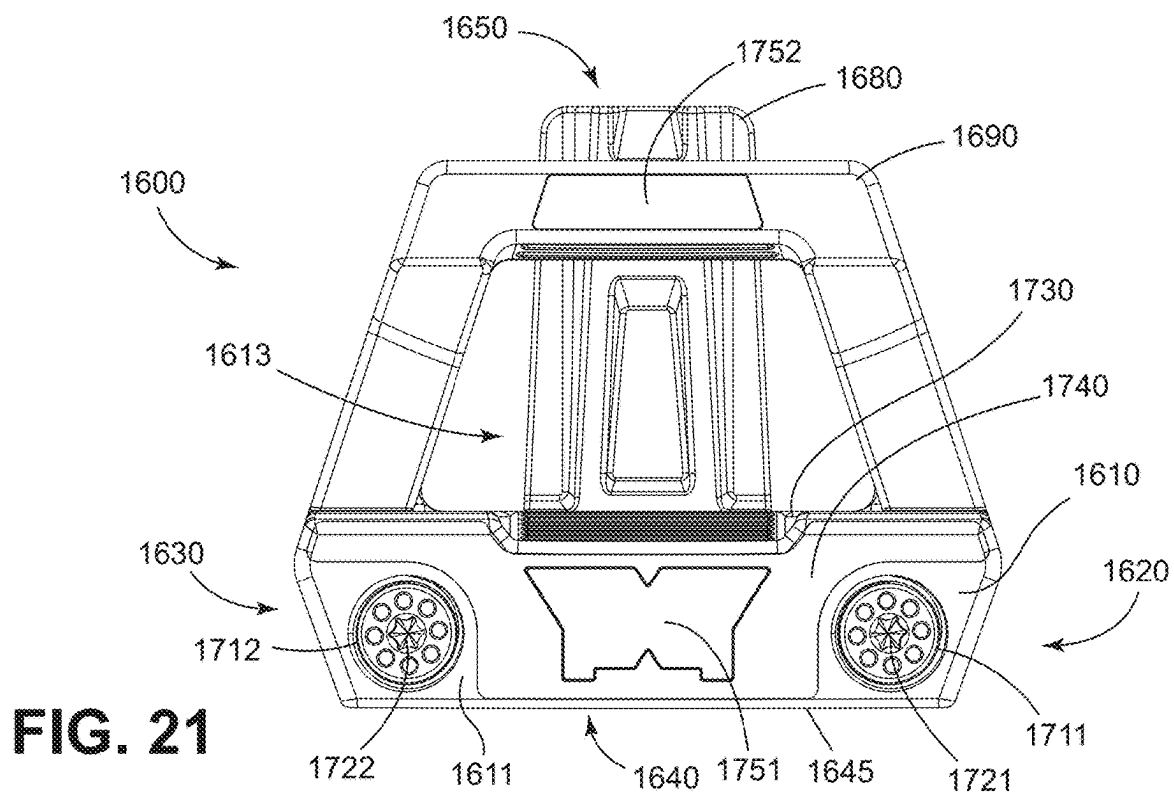
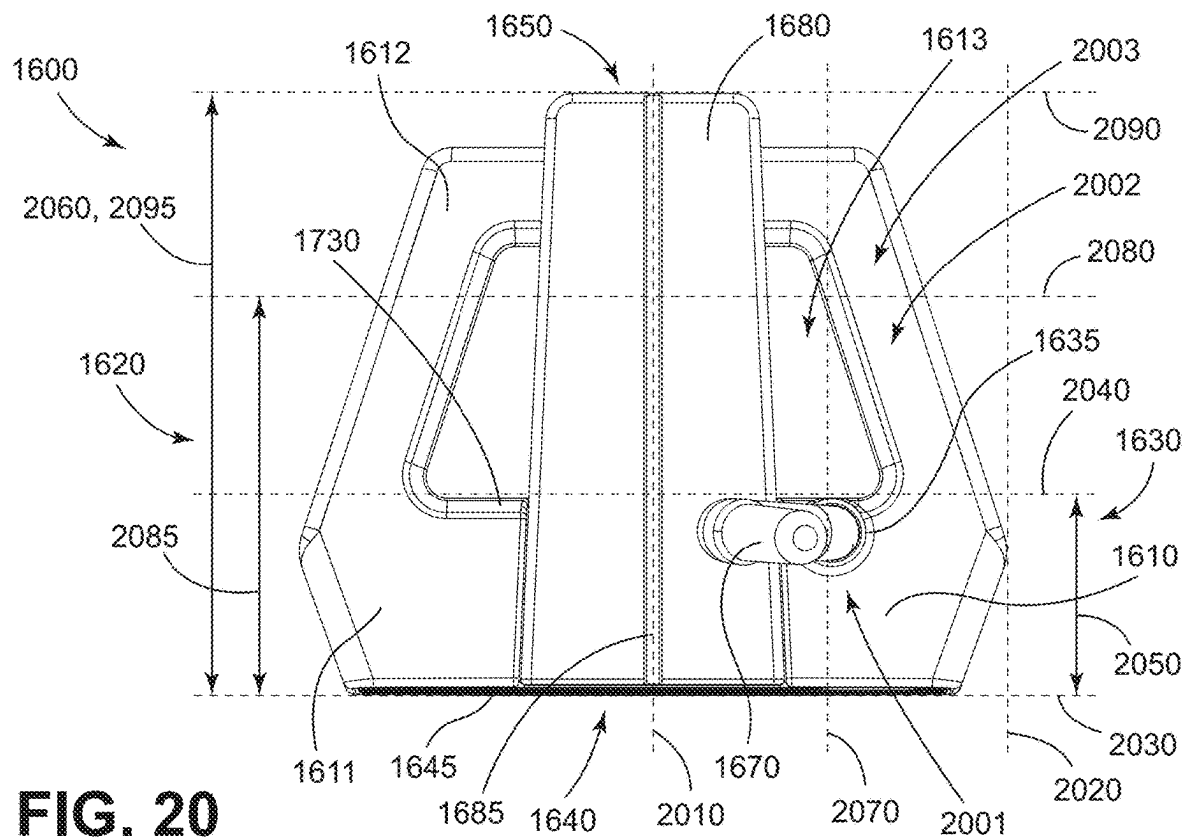
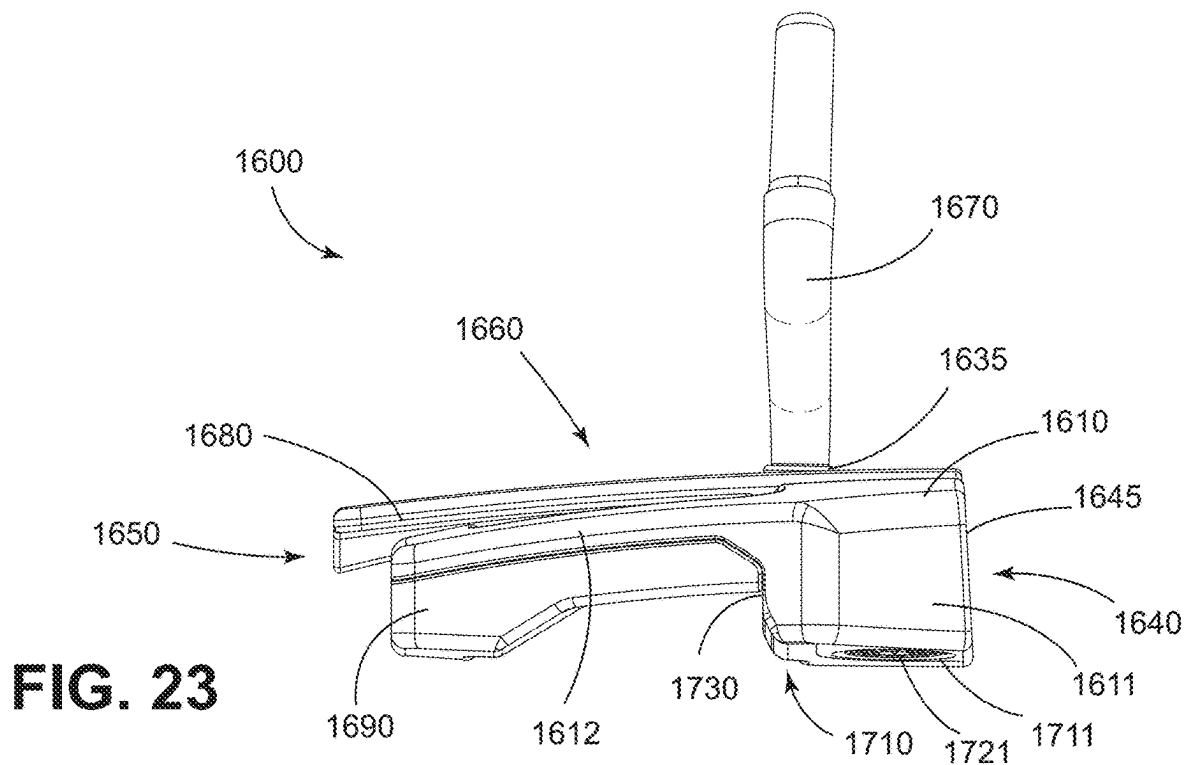
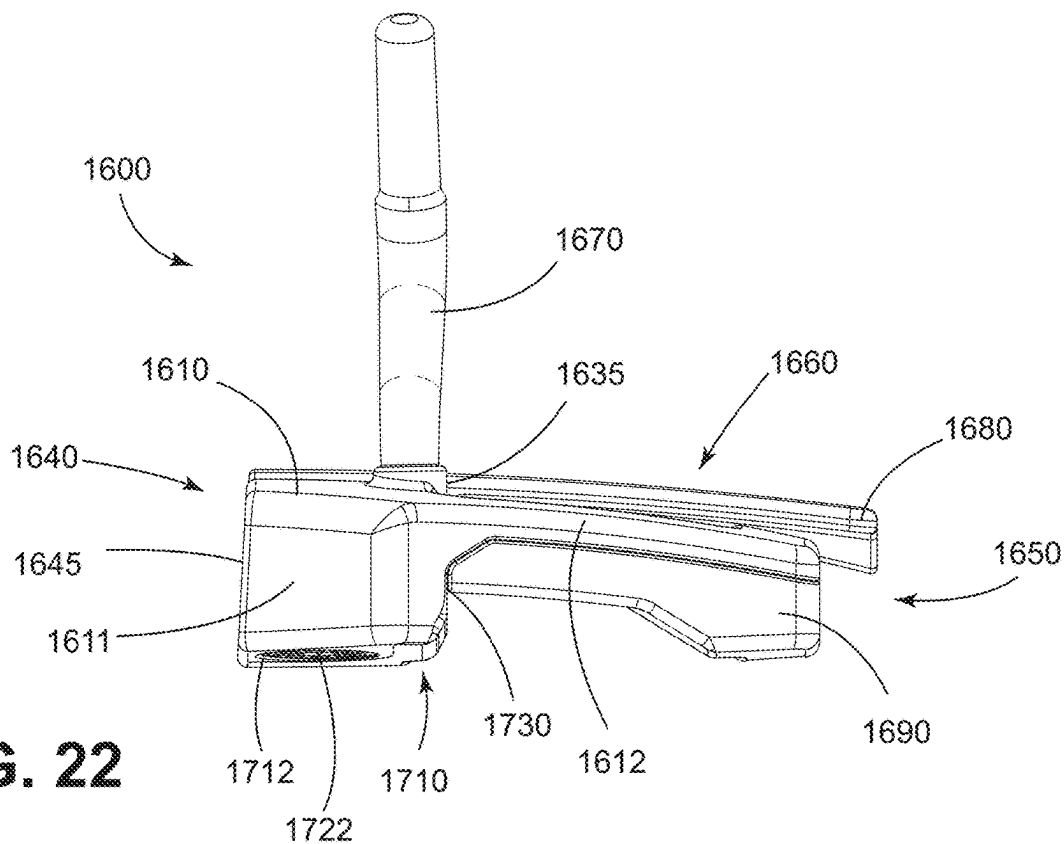


FIG. 19





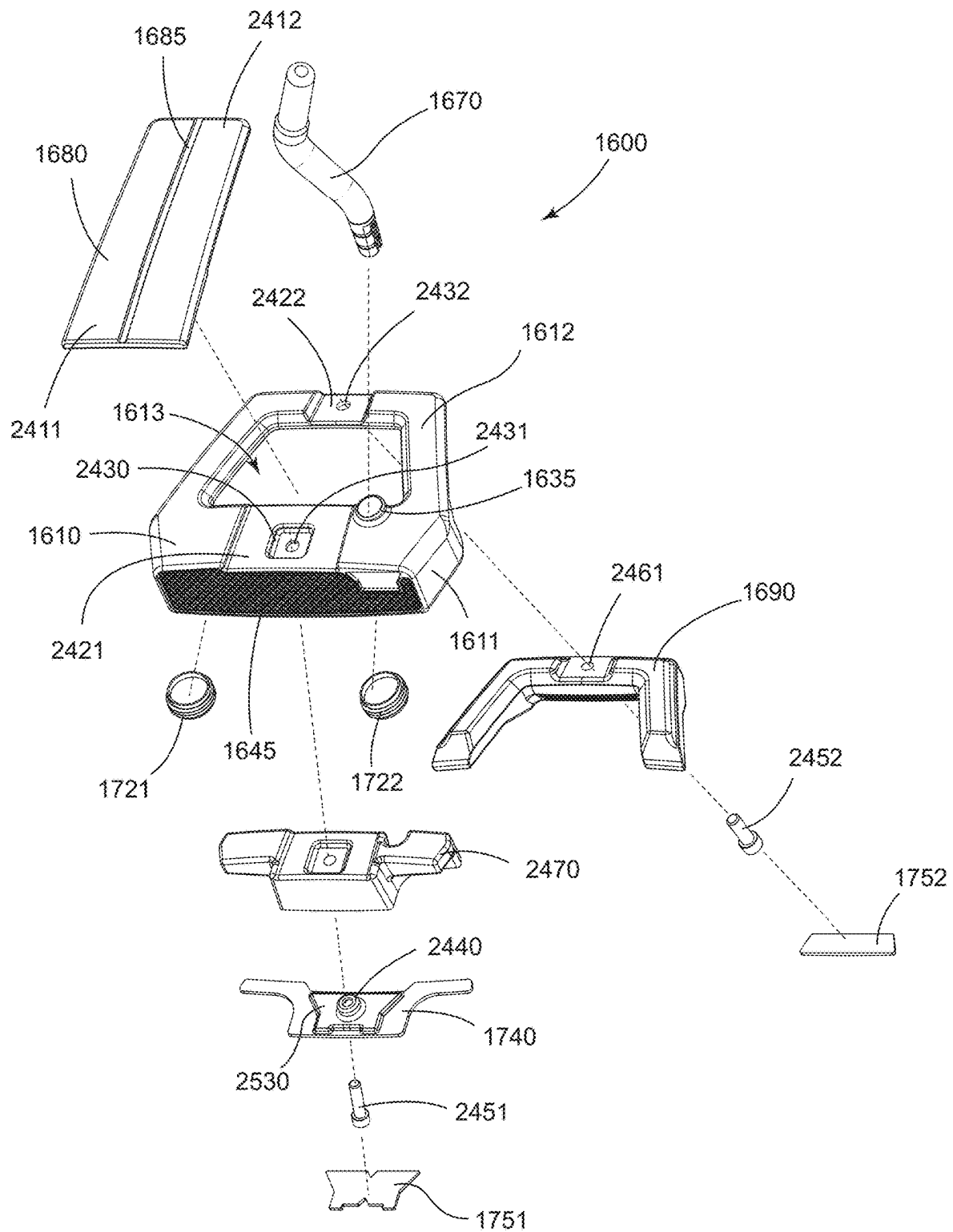


FIG. 24

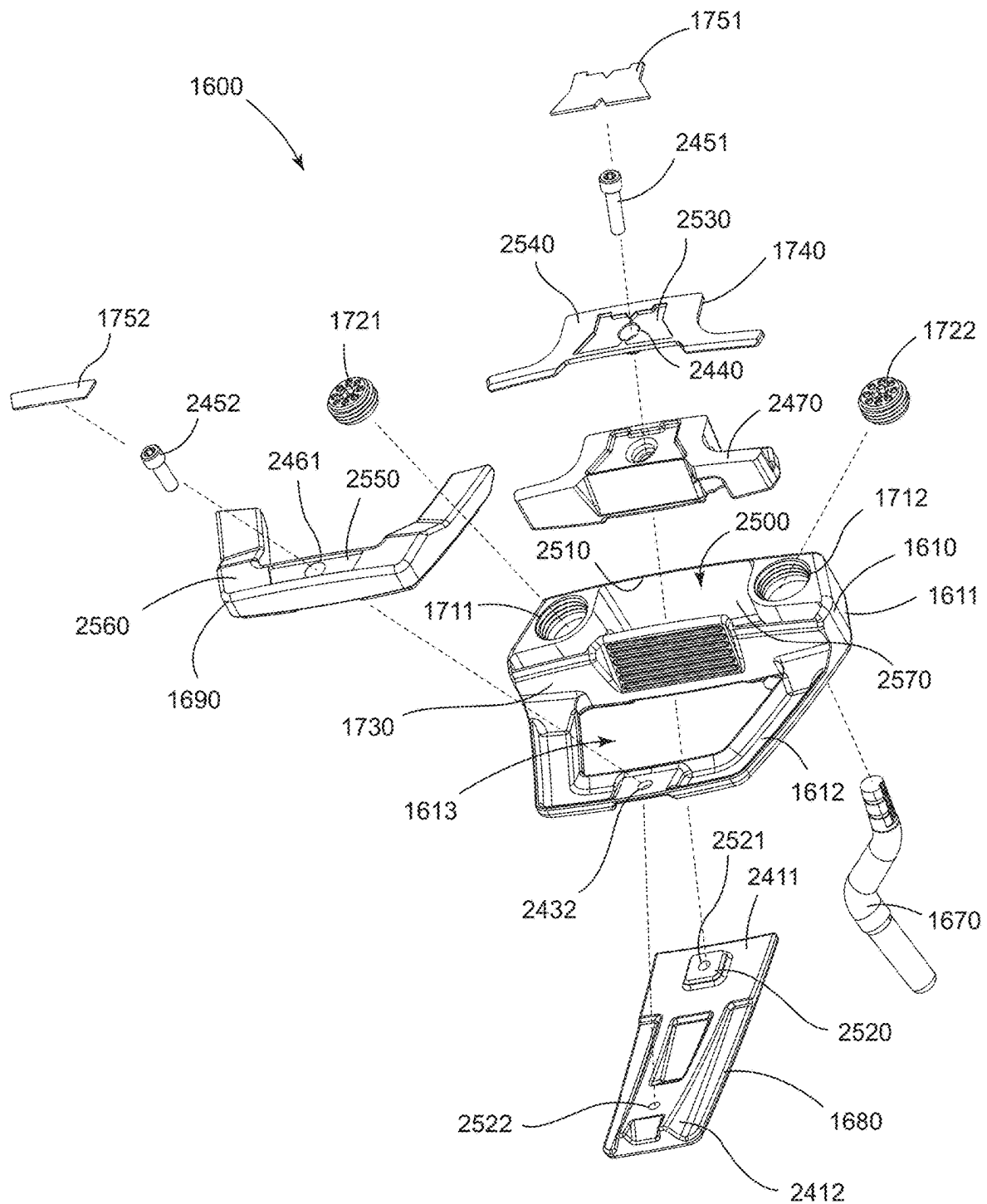


FIG. 25

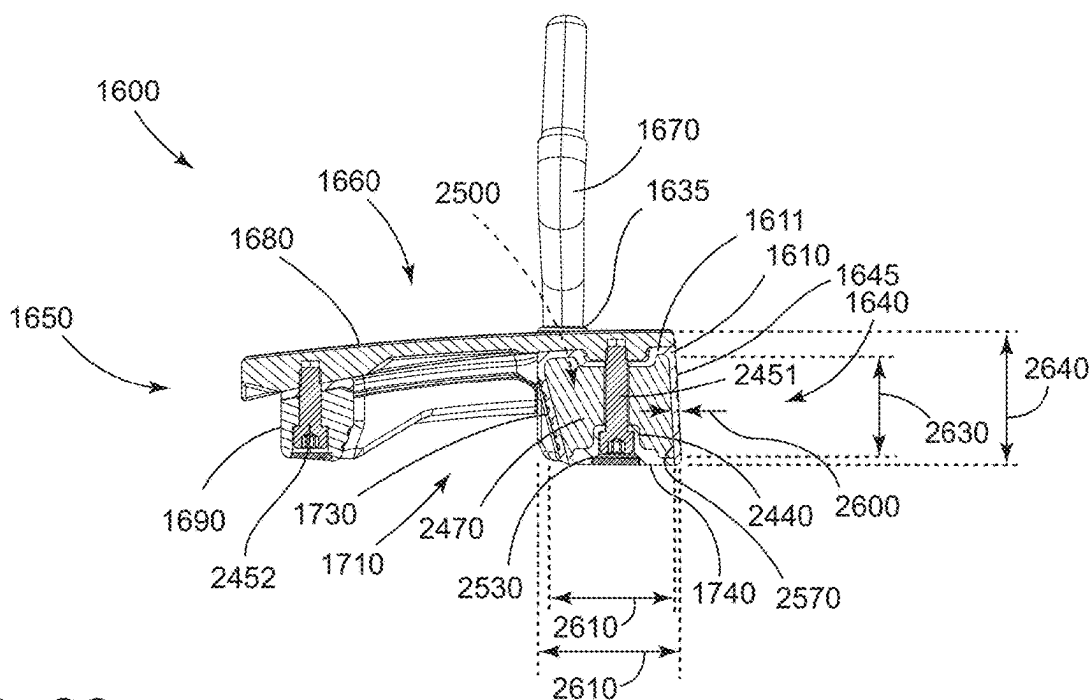


FIG. 26

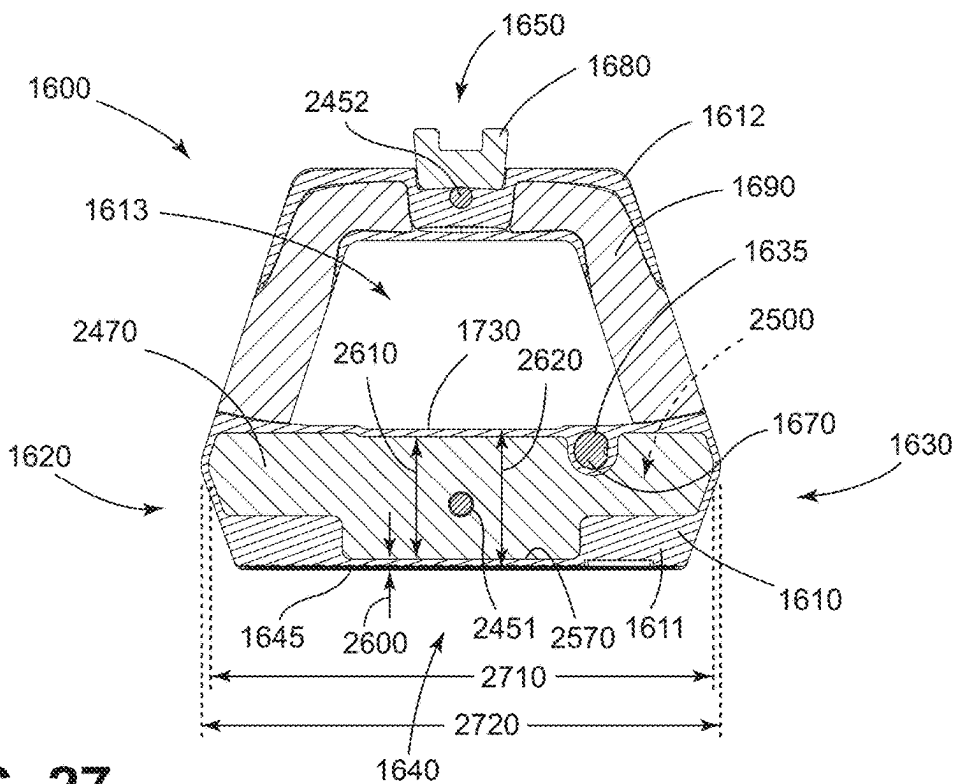


FIG. 27

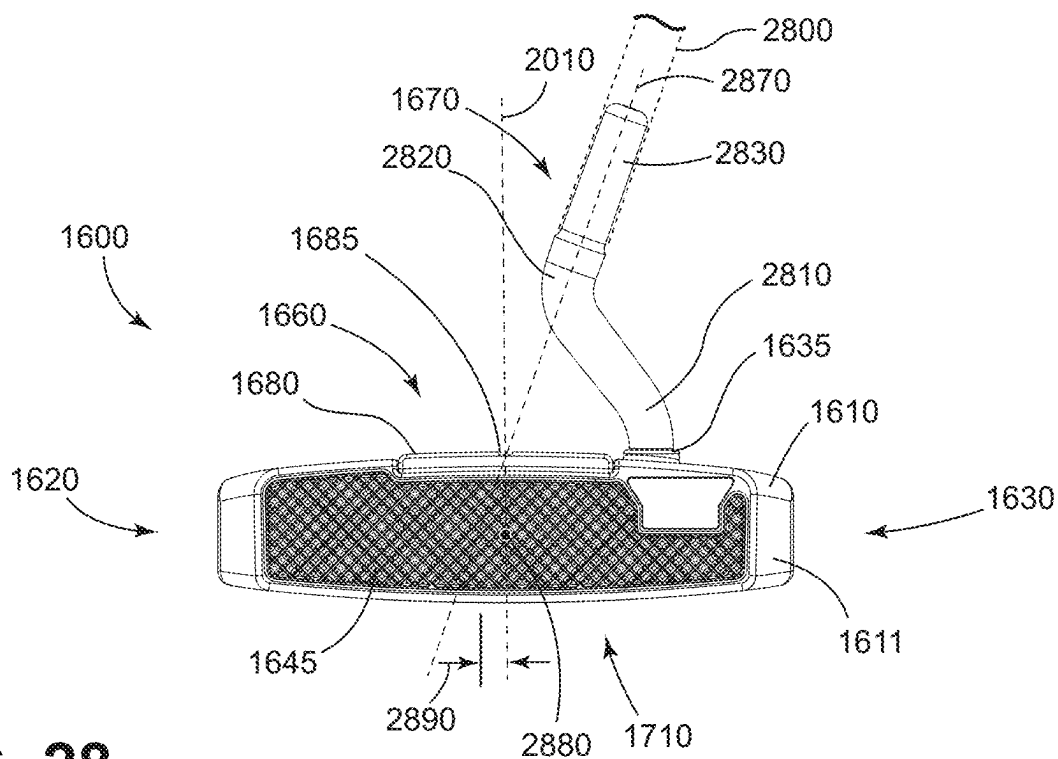


FIG. 28

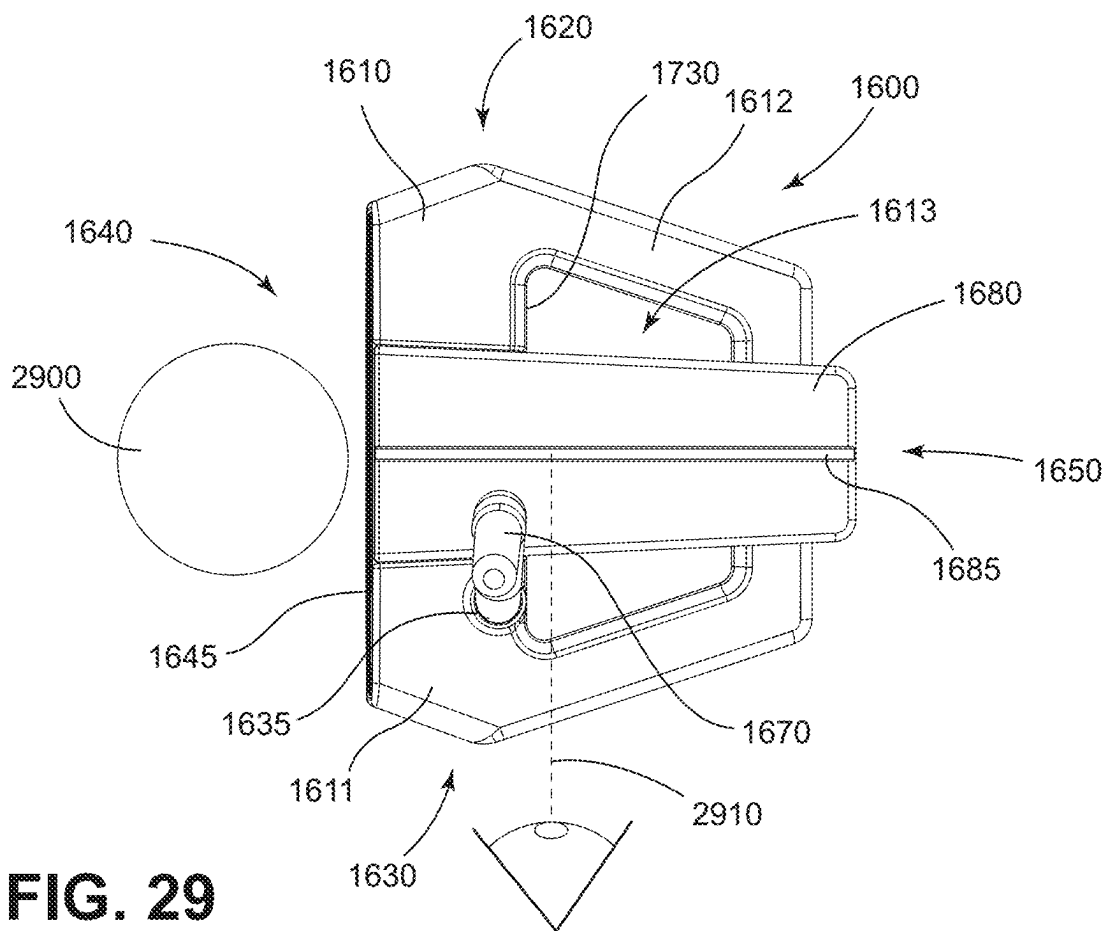


FIG. 29

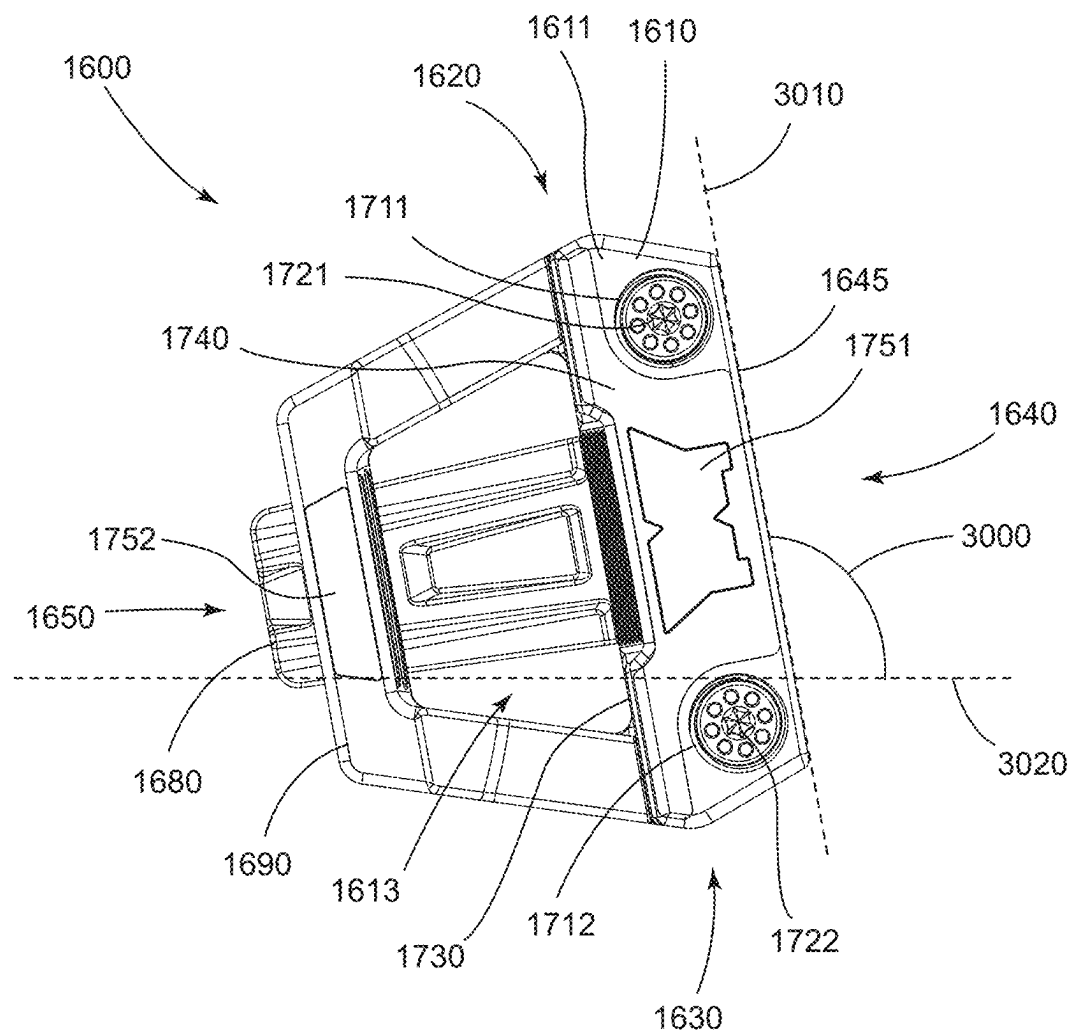
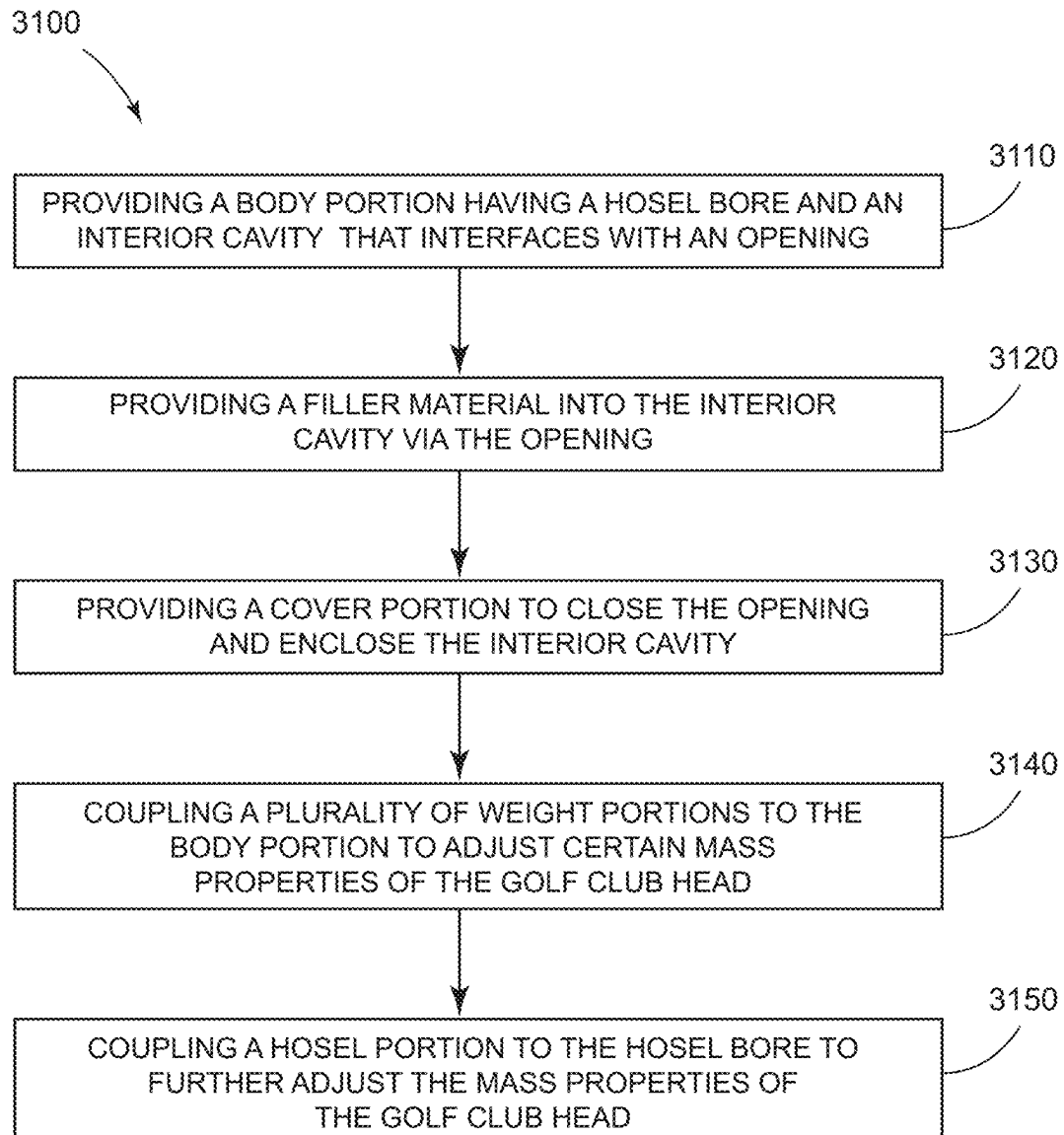
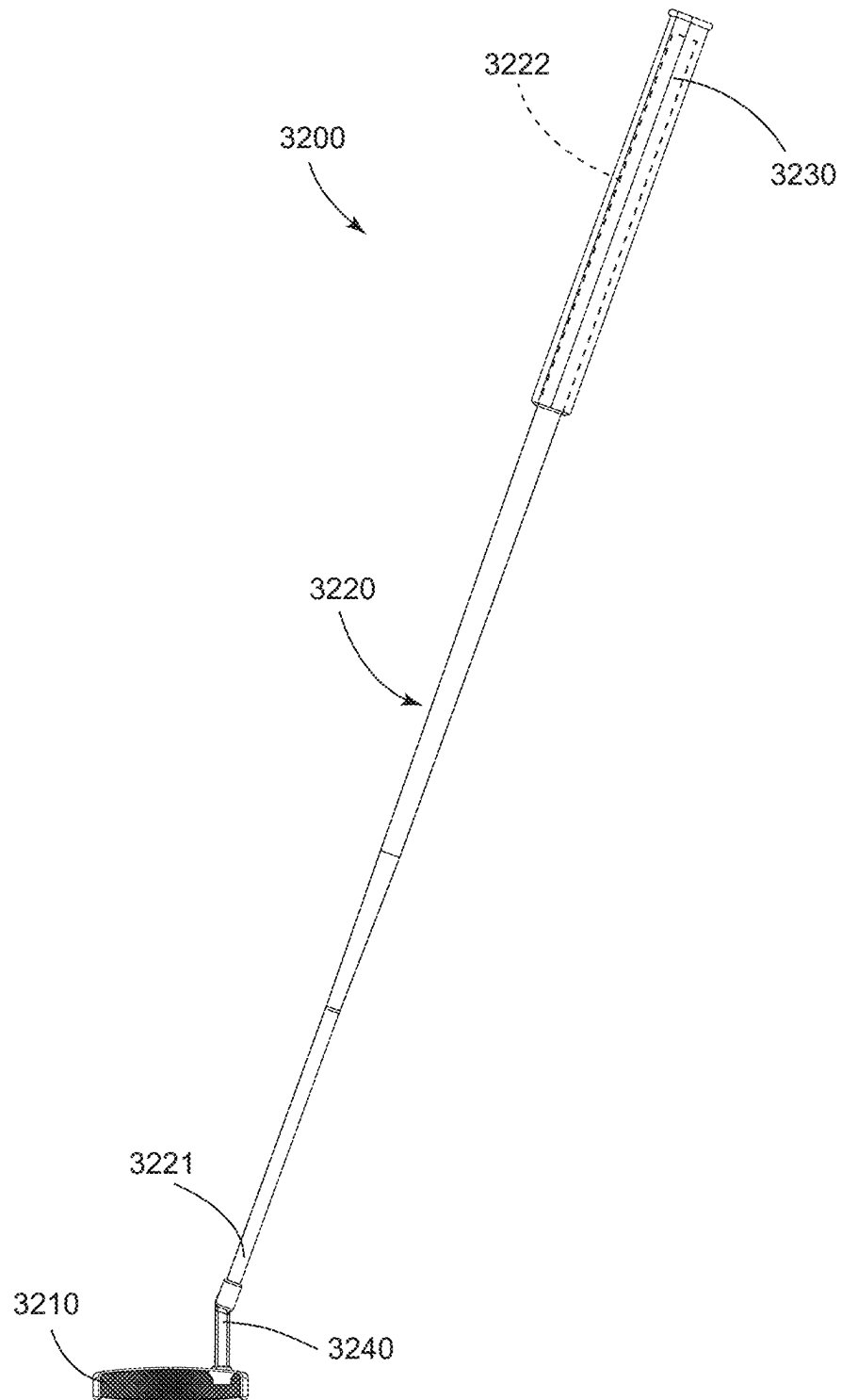
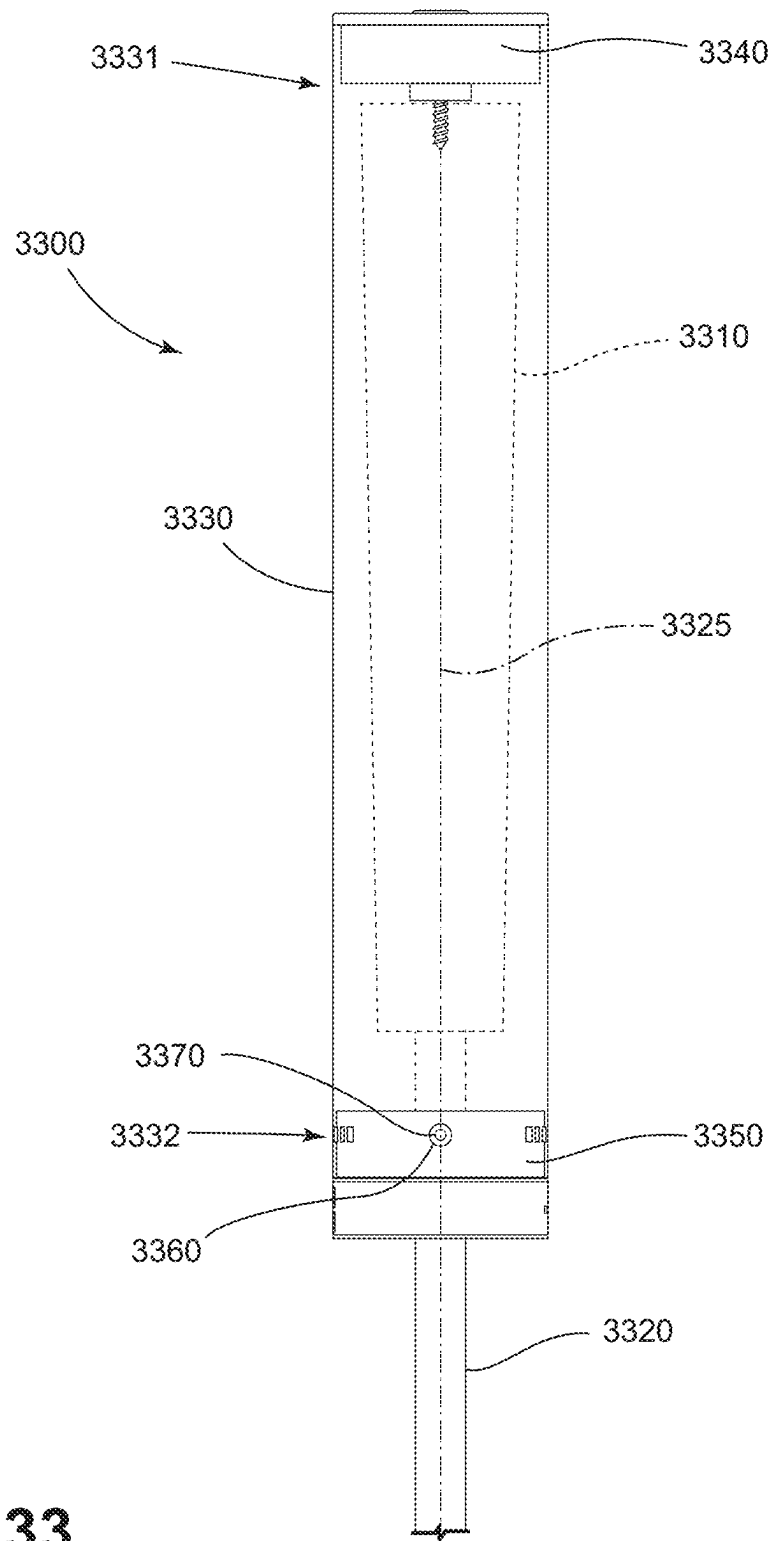


FIG. 30

**FIG. 31**

**FIG. 32**

**FIG. 33**

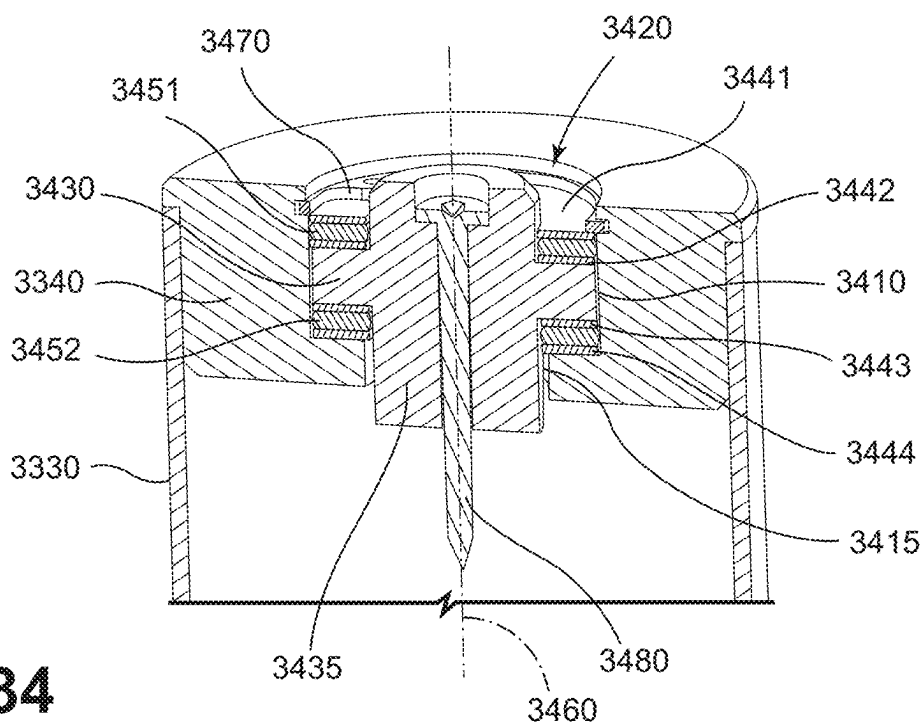


FIG. 34

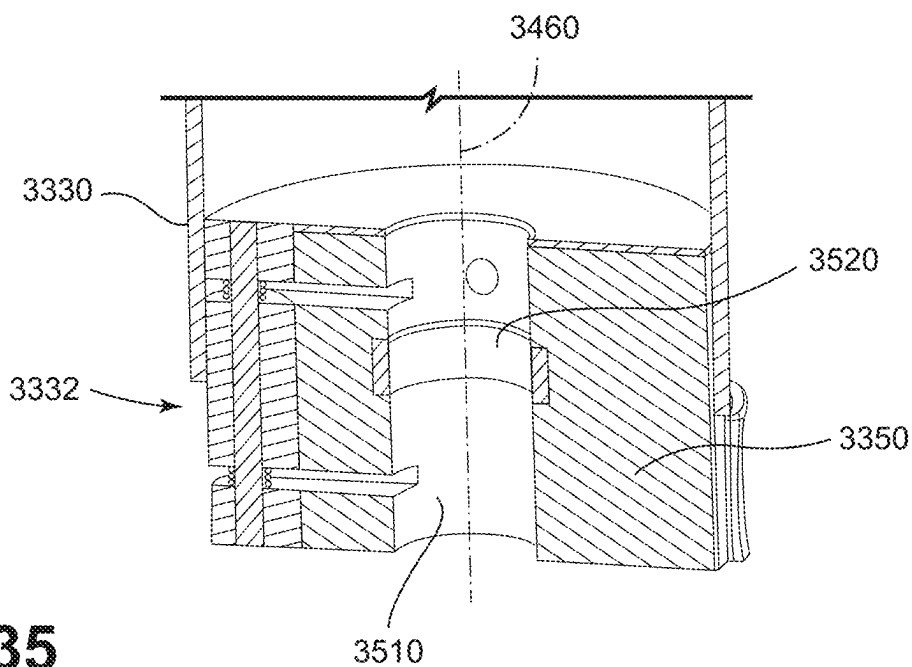


FIG. 35

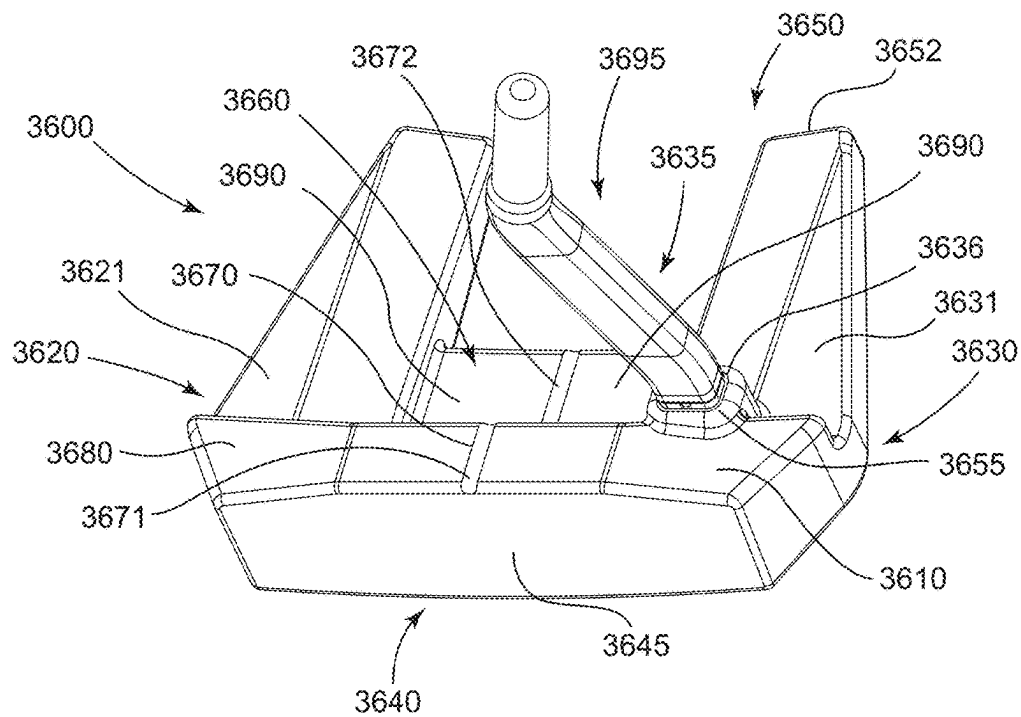


FIG. 36

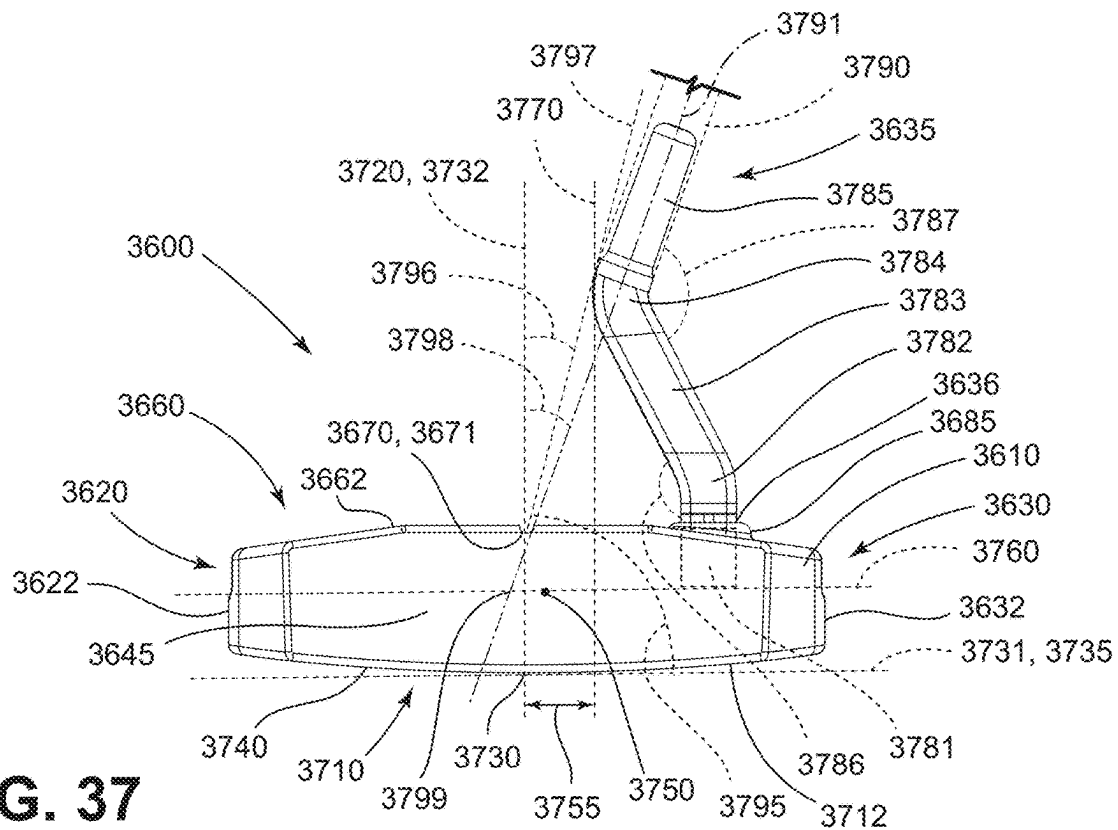


FIG. 37

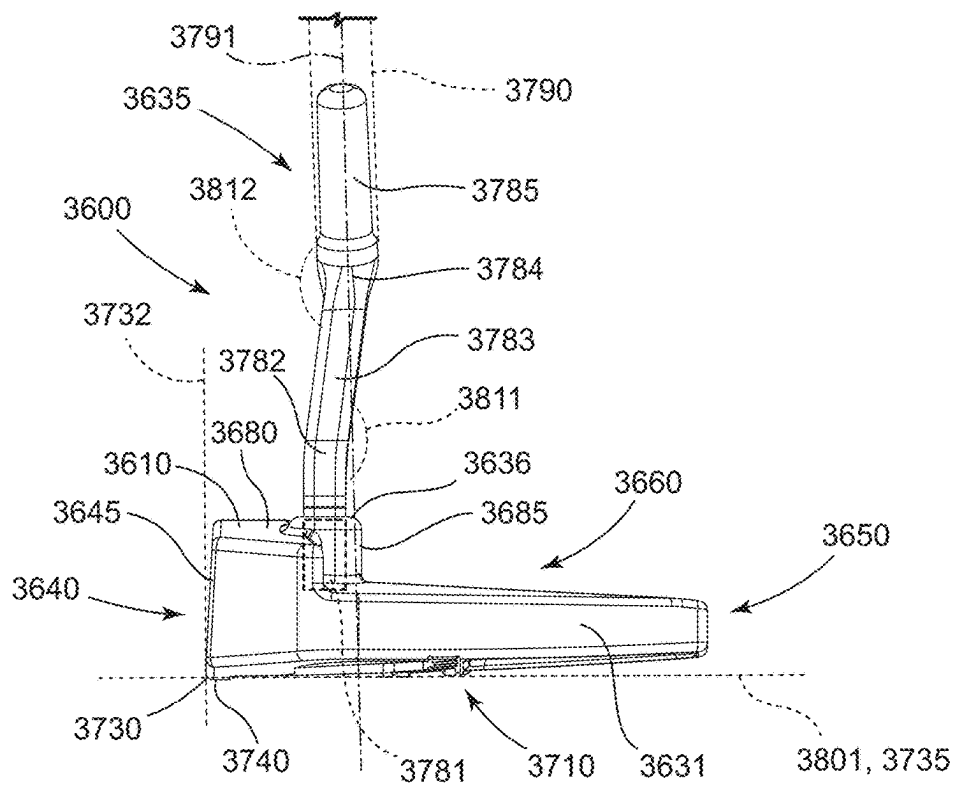


FIG. 38

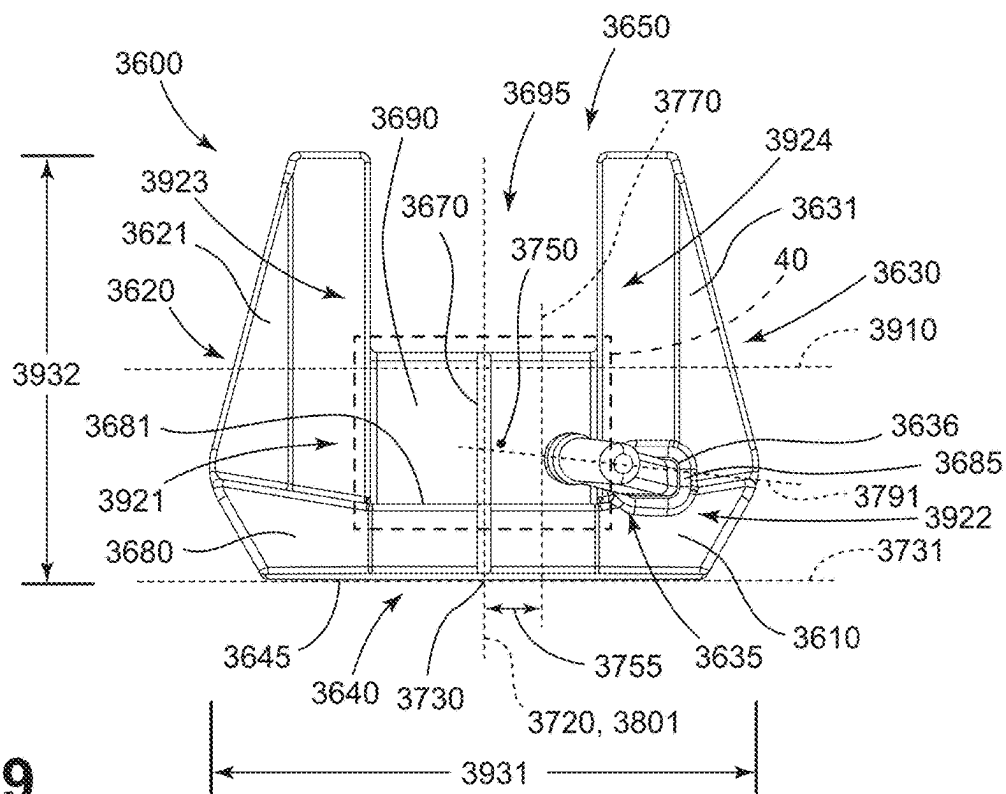
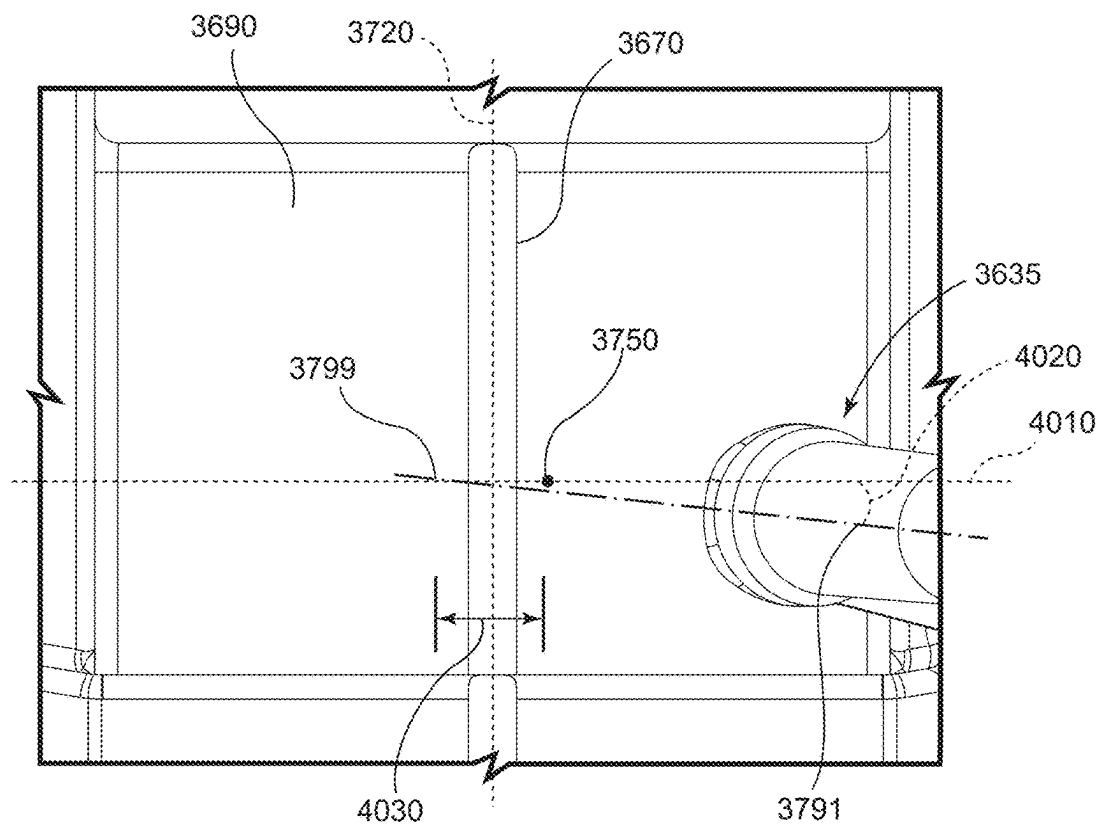


FIG. 39

**FIG. 40**

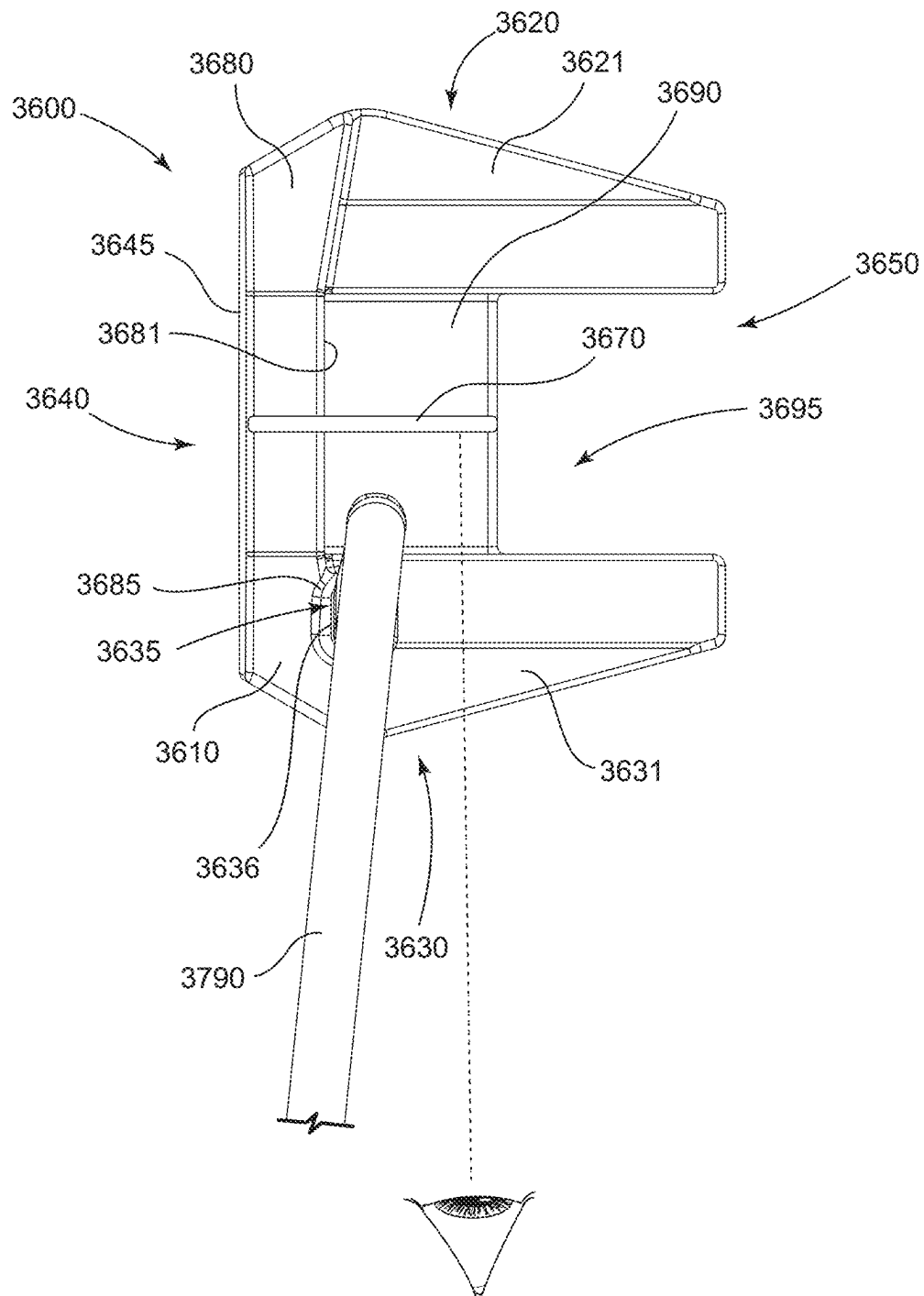
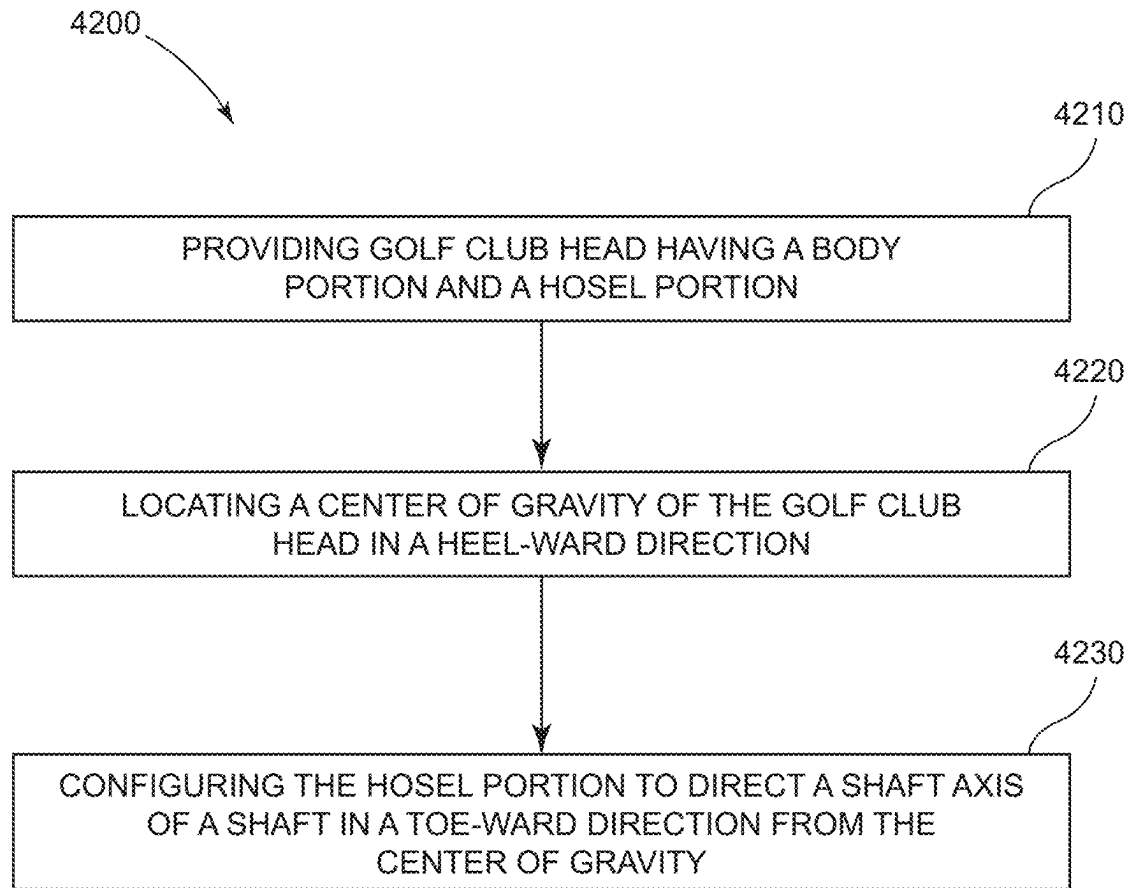


FIG. 41

**FIG. 42**

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GOLF CLUB HEADS AND METHODS TO MANUFACTURE GOLF CLUB HEADS

CROSS REFERENCE

This application claims the benefit of U.S. Provisional Application No. 63/708,955, filed Oct. 18, 2024.

This application claims the benefit of U.S. Provisional Application No. 63/672,521, filed Jul. 17, 2024.

This application is a continuation-in-part of application Ser. No. 18/423,608, filed Jan. 26, 2024, which is a continuation of application Ser. No. 18/241,633, filed Sep. 1, 2023, now U.S. Pat. No. 11,918,869, which claims the benefit of U.S. Provisional Application No. 63/525,847, filed Jul. 10, 2023.

U.S. application Ser. No. 18/241,633, filed Sep. 1, 2023, is a continuation-in-part of application Ser. No. 18/219,215, filed Jul. 7, 2023, now U.S. Pat. No. 11,839,801, which claims the benefit of U.S. Provisional Application No. 63/524,452, filed Jun. 30, 2023, and the benefit of U.S. Provisional Application No. 63/470,711, filed Jun. 2, 2023.

The disclosures of the above-referenced applications are incorporated by reference herein in their entirety.

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The present disclosure may be subject to copyright protection. The copyright owner has no objection to the facsimile reproduction by anyone of the present disclosure and its related documents, as they appear in the Patent and Trademark Office patent files or records, but otherwise reserves all applicable copyrights.

FIELD

The present disclosure generally relates to golf equipment, and more particularly, to golf club heads and methods to manufacture golf club heads.

BACKGROUND

Golf club heads may be configured with a variety of mass properties. By offsetting a center of gravity and a shaft axis, certain mass properties of a golf club head may be optimized to improve performance.

DESCRIPTION OF THE DRAWINGS

FIGS. 1, 2, 3, 4, 5, 6, 7, 8, and 9 depict a perspective front view, an exploded bottom perspective view, a front elevational view, a rear elevational view, an exploded rear perspective view, a perspective cross-sectional view (along line 6-6 of FIG. 4), a perspective cross-sectional view (along line 7-7 of FIG. 4), a perspective cross-sectional view (along line 8-8 of FIG. 4), and a perspective cross-sectional view (along line 9-9 of FIG. 4), respectively, of a golf club head according to an embodiment of the apparatus, methods, and articles of manufacture described herein.

FIG. 10 depicts an enlarged schematic front view of a face portion of the golf club head of FIGS. 1-9 according to an embodiment of the apparatus, methods, and articles of manufacture described herein.

FIG. 11 depicts a schematic cross-sectional view of a representative groove of the face portion of FIG. 10 according to an embodiment of the apparatus, methods, and articles of manufacture described herein.

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FIG. 12 depicts a schematic cross-sectional view of two representative grooves of the face portion of FIG. 10 according to an embodiment of the apparatus, methods, and articles of manufacture described herein.

FIGS. 13, 14, and 15 depict a front elevational view of the golf club head of FIGS. 1-9 assembled to a heel-shafted hosel type, a double bend hosel type, and a plumber's neck hosel type, respectively, according to an embodiment of the apparatus, methods, and articles of manufacture described herein.

FIGS. 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, and 27 depict a perspective front view, a perspective rear view, a front elevational view, a rear elevational view, a top view, a bottom view, a left view, a right view, an exploded top perspective view, an exploded bottom perspective view, a perspective cross-sectional view (along line 26-26 of FIG. 18), and a perspective cross-sectional view (along line 27-27 of FIG. 18), respectively, of a golf club head according to an embodiment of the apparatus, methods, and articles of manufacture described herein.

FIG. 28 depicts a shaft axis oriented toe-ward relative to a center of gravity of the golf club head of FIG. 16 according to an embodiment of the apparatus, methods, and articles of manufacture described herein.

FIG. 29 depicts the golf club head of FIG. 16 at an address position to strike a golf ball according to an embodiment of the apparatus, methods, and articles of manufacture described herein.

FIG. 30 depicts a hang angle of the golf club head of FIG. 16 according to an embodiment of the apparatus, methods, and articles of manufacture described herein.

FIG. 31 depicts a manner in which any of the example golf club heads described herein may be manufactured according to an embodiment of the apparatus, methods, and articles of manufacture described herein.

FIG. 32 depicts a golf club having a golf club head according to any embodiment of the apparatus, methods, and articles of manufacture described herein.

FIG. 33 depicts a grip attachment device according to an embodiment of the apparatus, methods, and articles of manufacture described herein.

FIG. 34 depicts a schematic cross-sectional view of an upper portion of the grip attachment device of FIG. 33.

FIG. 35 depicts a schematic cross-sectional view of a lower portion of the grip attachment device of FIG. 33.

FIG. 36 depicts a perspective front view of a golf club head according to an embodiment of the apparatus, methods, and articles of manufacture described herein.

FIG. 37 depicts a front view of the golf club head of FIG. 36.

FIG. 38 depicts a heel-side view of the golf club head of FIG. 36.

FIG. 39 depicts a top view of the golf club head of FIG. 36.

FIG. 40 depicts an enlarged view of an area of the golf club head designated by box 40 in FIG. 39.

FIG. 41 depicts the golf club head of FIG. 36 at an address position to strike a golf ball.

FIG. 42 depicts a manner in which any of the example golf club heads described herein may be manufactured according to an embodiment of the apparatus, methods, and articles of manufacture described herein.

For simplicity and clarity of illustration, the drawing figures illustrate the general manner of construction, and descriptions and details of well-known features and techniques may be omitted to avoid unnecessarily obscuring the present disclosure. Additionally, elements in the drawing

figures may not be depicted to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help improve understanding of embodiments of the present disclosure.

DESCRIPTION

The following U.S. Patents and Patent Applications, which are collectively referred to herein as “the incorporated by reference patent documents,” are incorporated by reference herein in their entirety: U.S. Pat. Nos. 9,233,283; 9,387,375; 9,440,124; 9,649,540; 9,895,585; 10,478,680; 10,493,331; 10,576,339; 10,737,153; 10,821,341; 10,960,271; 10,981,038; 11,045,698; 11,298,597; 11,369,849; and 11,517,798; and U.S. Patent Publication Nos. 20180200589 and 20220219054.

In general, golf club heads and methods to manufacture golf club heads are described herein. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

In the example of FIGS. 1-9, a golf club head **100** is generally shown as a first type of blade putter. The golf club head **100** may include a body portion **110** made from any of the materials described herein or described in any of the incorporated by reference patent documents. The body portion **110** may have a toe portion **120**, a heel portion **130** opposite the toe portion **120**, a hosel bore **135** located at the heel portion **130**, a front portion **140**, a face portion **145** located at the front portion **140**, a rear portion **150** opposite the front portion **140**, a top portion **160**, and a sole portion **210** opposite the top portion **160**. The toe portion **120** may include a toe portion edge **122**, the heel portion **130** may include a heel portion edge **132**, the rear portion **150** may include a rear portion edge **152**, the top portion **160** may include a top portion edge **162**, and the sole portion **210** may include a sole portion edge **212**. The body portion **110** may be manufactured via various manufacturing methods and/or processes (e.g., a casting process, a forging process, a milling process, a cutting process, a grinding process, a welding process, a 3D printing process, a combination thereof, etc.). The body portion **110** may be partially or entirely made of an aluminum-based material (e.g., a high-strength aluminum alloy or a composite aluminum alloy coated with a high-strength alloy), a magnesium-based material, a stainless steel-based material, a titanium-based material, a tungsten-based material, any combination thereof, and/or other suitable types of materials. Alternatively, the body portion **110** may be partially or entirely made of a non-metal material (e.g., composite, plastic, etc.). The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

The face portion **145** may define a front surface of the body portion **110** for striking a golf ball. The face portion **145** may be generally planar and may have a loft angle greater than or equal to two degrees and less than or equal to four degrees. The sole portion **210** may include a plurality of weight ports (e.g., shown as a toe-side weight port **220** and a heel-side weight port **230**) each configured to receive a complementary weight portion (e.g., shown as weight portions **240** and **250**). Weight portions **240** and **250** may be configured as interchangeable screws threadedly coupled to the toe-side weight port **220** and the heel-side weight port **230**, respectively. Weight portions **240** and **250** may be made from any of the materials described herein and may have similar or different material compositions and/or physical properties (e.g., density, color, texture) with respect to each other and the body portion **110**. The hosel bore **135** may be

configured to receive a variety of hosel types to accommodate a variety of swing types. The configuration and/or properties of each of the weight ports and the weight portions may be similar in many respects to any of the weight ports and weight portions, respectively, described in any of the incorporated by reference patent documents. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

The golf club head **100** may include an interior cavity **500** within the body portion **110**. The interior cavity **500** may be located between the face portion **145** and a back wall portion **400** disposed opposite the face portion **145**. The back wall portion **400** may extend between the toe portion **120** and the heel portion **130** and may include a recessed portion **510** defining an opening **520** that interfaces with the interior cavity **500**. The recessed portion **510** may be located adjacent a ledge portion **530** at or proximate the rear portion **150** and extending between a toe-side mass platform **540** and a heel-side mass platform **550**. In the example of FIGS. 1-9, the ledge portion **530** may be recessed relative to the toe-side mass platform **540** and the heel-side mass platform **550**. The interior cavity **500** may be partially or entirely filled with a filler material **560**. The filler material **560** may be any type of polymer material described herein or described in any of the incorporated by reference patent documents. In one example, the filler material **560** may include a low-density polymer material. In another example, the filler material **560** may be a solid piece manufactured in the shape of the interior cavity **500** so as to be insertable into the interior cavity **500** through the opening **520**. In yet another example, the filler material **560** may be an injectable liquid or fluid material provided to the interior cavity **500** via the opening **520** and later hardened at ambient conditions or during a curing process. The golf club head **100** may include a cover portion shown for example as back plate **410** that may be coupled to the body portion **110** to close the opening **520** and to enclose the interior cavity **500**. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

The back plate **410** may have a complementary shape to the recessed portion **510**. The back plate **410** may include a front surface **571**, a rear surface **572**, and a stiffening member **573**. In one example, as illustrated in FIGS. 1-9, the stiffening member **573** may be configured as an elongated rib or protrusion extending longitudinally across the front surface **571** of the back plate **410**. The back plate **410** may be joined to the body portion **110** such that the front surface **571** of the back plate **410** contacts the recessed portion **510** and the end portions of the stiffening member **573** are each received in a complementary indent (e.g., shown as indents **511** and **512**) in the recessed portion **510**. In the present example, the stiffening member **573** may extend across the opening **520** in a toe-to-heel direction of the golf club head **100**. Indents **511** and **512** may each include a fastener port (e.g., shown as fastener ports **513** and **514**) concentrically aligned with a complementary through-port (e.g., shown as through-ports **574** and **575**) extending through the rear surface **572** and the stiffening member **573** of the back plate **410**. The back plate **410** may be attached to the body portion **110** via a first mechanical fastener **421** inserted in through-port **574** and threadedly engaged to fastener port **513** and a second mechanical fastener **422** inserted in through-port **575** and threadedly engaged to fastener port **514**. In another example, only an adhesive may be used to join the front surface **571** of the back plate **410** to the recessed portion **510**. In yet another example, fasteners and one or more adhesives may be used to join the front surface **571** of the back plate

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410 to the recessed portion 510. In operation, the stiffening member 573 may impart structural rigidity to the back plate 410 and may also compress the filler material 560 against a rear surface 590 of the face portion 145. The back plate 410 may be made from any of the materials described herein or described in any of the incorporated by reference patent documents. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

The filler material 560 may contact and reinforce the rear surface 590 of the face portion 145 to reduce face deformation and also dampen sound and vibration. The filler material 560 may also encase a portion of the stiffening member 573 extending across the opening 520 to help maintain the back plate 410 in place. The rear surface 590 of the face portion 145 may include a rear surface of a central strike portion 700 of the face portion 145. The central strike portion 700 may correspond to a portion of the face portion 145 bisected by a center longitudinal plane 900 of the golf club head 100 and may extend a length 910 between the toe portion 120 and the heel portion 130. In one example, the length 910 of the central strike portion 700 may be greater than or equal to 1.00 inch (25.4 mm) and less than or equal to 1.70 inch (43.18 mm). In another example, the length 910 of the central strike portion 700 may be greater than or equal to 1.20 inches (30.48 mm) and less than or equal to 1.70 inch (43.18 mm). In yet another example, the length 910 of the central strike portion 700 may be greater than or equal to 1.40 inches (35.56 mm) and less than or equal to 1.70 inch (43.18 mm). The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

In one example, as illustrated in FIG. 7, the golf club head 100 may have at least one cross-section where the filler material 560 has a length 710 in a front-to-rear direction greater than or equal to 40% of a length 720 between the front surface of the face portion 145 and the rear surface 572 of the back plate 410. With continued reference to FIG. 7, the golf club head 100 may have at least one cross-section where the filler material 560 has a length 730 in a top-to-sole direction greater than or equal to 70% of a length 740 of the body portion 110 in the top-to-sole direction. With reference to FIG. 9, the golf club head 100 may have at least one cross-section where the filler material 560 has a length 920 in a toe-to-heel direction greater than or equal to the length 910 of the central strike portion 700 and/or greater than or equal to 30% of a length 930 of the body portion 110 in the toe-to-heel direction. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

The filler material 560 may have a density less than a density of the body portion 110 so that more discretionary mass may be allocated to other areas of the golf club head 100. With respect to the present example, the filler material 560 may have a fill volume (V_f) greater than or equal to 6% and less than or equal to 10% of a total volume (V_t) of the golf club head 100 and a fill mass (M_f) greater than or equal to 0.5% and less than or equal to 1% of a total mass (M_t) of the golf club head 100. In one example, the total volume (V_t) of the golf club head 100 may be greater than or equal to 2.50 inch³ (40.968 cm³) and less than or equal to 3.00 inch³ (49.161 cm³) and the total mass (M_t) of the golf club head 100 may be greater than or equal to 300 grams and less than or equal to 345 grams. Selecting the filler material 560 from a lightweight or low-density material and increasing the fill volume (V_f) of the filler material 560 relative to the total volume (V_t) of the golf club head 100 may enable greater discretionary mass placement toward a periphery of the golf club head 100 to optimize certain mass properties of the golf club head 100 such as moment of inertia (MOI), center of

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gravity (CG) location, and mass balance while maintaining a suitable club head weight. With respect to any of the examples described herein, a periphery of a golf club head may include the perimeter portions of the toe portion, the heel portion, and the rear portion, respectively. As defined herein, a ratio between a percentage fill volume

$$\left(\frac{V_f}{V_t} \times 100\right)$$

and a percentage of fill mass

$$\left(\frac{M_f}{M_t} \times 100\right)$$

of a filler material may be indicative of the level of freedom in which discretionary mass may be distributed to other areas of a golf club head. Said differently, the ratio between a percentage fill volume and a percentage fill mass of a filler material may correspond to the percentage fill volume divided by the percentage fill mass and a larger ratio value may signify a greater degree of freedom in discretionary mass placement. In one example, the filler material 560 may have a ratio between a percentage fill volume and a percentage fill mass of a filler material that may be greater than or equal to 6.00 and the golf club head 100 may have a MOI greater than or equal to 4500 g·cm². In another example, the golf club head 100 may have a MOI greater than or equal to 4700 g·cm². In yet another example, the golf club head 100 may have a MOI greater than or equal to 5000 g·cm². The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

The golf club head 100 may have thin cavity walls to provide additional discretionary mass to be allocated to other areas of the golf club head 100 compared to conventional solid-body golf club heads. For example, one or more cavity walls may have a thickness (constant or average) greater than or equal to 0.035 inch (0.889 mm) and less than or equal to 0.060 inch (1.524 mm). In particular, the face portion 145 may define a thin front cavity wall reinforced by the filler material 560. Accordingly, the structural support provided by the filler material 560 may allow for having relatively thinner face portion and cavity walls. With reference to FIG. 10, the face portion 145 may include a plurality of grooves 1000 defined by a perimeter groove 1010, a first plurality of grooves (e.g., shown as grooves 1021, 1022, and 1023), and a second plurality of grooves (e.g., shown as grooves 1031, 1032, and 1033). In the example of FIG. 10, dashed lines have been overlaid onto the perimeter groove 1010 and the example grooves for ease of reference and purposes of understanding. In the illustrated example, the perimeter groove 1010 may define a boundary enclosing the first plurality of grooves and the second plurality of grooves. The first plurality of grooves may be arranged in parallel with one another and may extend in a first diagonal direction across face portion 145 within the boundary set by the perimeter groove 1010. The second plurality of grooves may be arranged in parallel with one another and may extend in a second diagonal direction within the boundary set by the perimeter groove 1010. Each groove of the first plurality of grooves and the second plurality of grooves may terminate at the perimeter groove 1010. In other words, the first plurality of grooves and the second plurality of grooves may be joined to the perimeter groove 1010 at their terminal

ends. In one example, the first plurality of grooves may be transverse to the second plurality of grooves. Accordingly, the first plurality of grooves may intersect with the second plurality of grooves at right angles. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

With continued reference to FIG. 10, the plurality of grooves **1000** may define a first plurality of projections (e.g., shown as projections **1041**, **1042**, and **1043**) and a second plurality of projections (e.g., shown as projections **1051**, **1052**, and **1053**). Each projection (e.g., projection **1041**) of the first plurality of projections may be disposed inward of the perimeter groove **1010** and may correspond to a raised structure occupying an intersectional space between two adjacent grooves (e.g., grooves **1021** and **1022**) of the first plurality of grooves and two adjacent grooves (e.g., grooves **1031** and **1032**) of the second plurality of grooves. In other words, each projection of the first plurality of projections may be bounded by two adjacent grooves of the first plurality of grooves and two adjacent grooves of the second plurality of grooves intersecting the two adjacent grooves of the first plurality of grooves. In one example, each projection of the first plurality of projections may have a frusto-pyramidal shape with a flat peak surface for striking a golf ball. Each projection (e.g., projection **1051**) of the second plurality of projections may be disposed adjacent the perimeter groove **1010** and may correspond to a raised structure occupying an intersectional space between the perimeter groove **1010**, at least one groove (e.g., groove **1023**) of the first plurality of grooves, and at least one groove (e.g., groove **1031**) of the second plurality of grooves. In other words, each projection of the second plurality of projections may be bounded by the perimeter groove **1010**, at least one groove of the first plurality of grooves, and at least one groove of the second plurality of grooves intersecting the at least one groove of the first plurality of grooves. Each projection of the second plurality of projections may have a variety of shapes based on the intersectional relationship between the perimeter groove **1010**, the first plurality of grooves, and the second plurality of grooves. In one example, each projection of the second plurality of projections may correspond to a partial segment of a projection of the first plurality of projections. While the examples and figures described herein may describe and depict a face portion having particular projections, the apparatus, methods, and articles of manufacture described herein may include projections of other suitable shapes, size, position, etc. For example, the face portion **145** may include at two projections with different shape or size. Alternatively, the face portion **145** may not include a projection. In one example, the face portion **145** may include one or more horizontal grooves extending between the toe portion **120** and the heel portion **130**. In another one example, the face portion **145** may include a smooth flat surface or a smooth contoured surface. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

The first plurality of grooves and the second plurality of grooves may have similar groove properties (e.g., width and depth) and may be evenly spaced apart such that the first plurality of projections are the same size and shape. For purposes of understanding, an example groove **1100** is shown in FIG. 11 and may be representative of any groove of the first plurality of grooves and the second plurality of grooves. In the present example, the groove **1100** may have a depth **1110** greater than or equal to 0.010 inch (0.254 mm) and less than or equal to 0.015 inch (0.381 mm). The groove **1100** may have a variable width that decreases at a first rate

from an upper width **1120** at the top of the groove **1100** to an intermediate width **1130** and decreases at a second rate from the intermediate width **1130** to a lower width **1140** at a base **1150** of the groove **1100**. In one example, the upper width **1120** may be greater than or equal to 0.045 inch (1.143 mm) and less than or equal to 0.060 inch (1.524 mm), the intermediate width **1130** may be greater than or equal to 0.015 inch (0.381 mm) and less than or equal to 0.020 inch (0.508 mm), and the lower width may be greater than or equal to 0.009 inch (0.2286 mm) and less than or equal to 0.012 inch (0.3048 mm). Additionally, a rate of decrease from the upper width **1120** to the intermediate width **1130** may occur linearly over a depth **1160** greater than or equal to 0.007 inch (0.1778 mm) and less than or equal to 0.010 inch (0.254 mm) and a rate of decrease from the intermediate width **1130** to the lower width **1140** may occur non-linearly over a depth **1170** greater than or equal 0.003 inch (0.0762 mm) and less than or equal to 0.005 inch (0.127 mm). In the example of FIG. 11, the groove **1100** may be characterized as having an upper cross-section **1180** with an inverted frustoconical shape and a lower cross-section **1190** with an inverted truncated funnel shape. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

Referring to FIG. 12, two adjacent grooves are shown as grooves **1210** and **1220** and may be representative of any two adjacent grooves of the first plurality of grooves or any two adjacent grooves of the second plurality of grooves. In the present example, a centerline spacing **1230** between grooves **1210** and **1220** may be greater than or equal to 0.060 inch (1.524 mm) and less than or equal to 0.100 inch (2.54 mm). In the present example, the face portion **145** may have a first face thickness **1240** greater than or equal to 0.035 inch (0.889 mm) and less than or equal to 0.055 inch (1.397 mm). The first face thickness **1240** may correspond to a distance between the rear surface **590** of the face portion **145** and a strike surface **1250** of the face portion **145**, which may include the flat peak surfaces of the first plurality of projections. In one example, the flat peak surfaces of the first plurality of projections may be coplanar. The face portion **145** may also have a second face thickness **1260** greater than or equal to 0.035 inch (0.889 mm) and less than or equal to 0.045 inch (1.143 mm). The second face thickness **1260** may correspond to a distance between the rear surface **590** of the face portion **145** and a base (e.g., shown as base **1270**) of a groove (e.g., groove **1210**). In one example, the bases of the first plurality of grooves and the second plurality of grooves may be coplanar. The face portion **145** incorporating the plurality of grooves **1000** described herein may cooperate with the filler material **560** to provide enhanced performance properties such as consistent roll distance and improved feel and sound. The configurations and properties of the face portion and the plurality of grooves may be similar in many respects to the face portion and the plurality of the grooves described in U.S. Pat. Nos. 10,737,153 and 10,981,038, which are incorporated by reference herein. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

The mass properties of the golf club head **100** may be adjusted using a weighting system that may utilize a plurality of weight ports (e.g., toe-side weight port **220** and heel-side weight port **230**) and corresponding interchangeable weight portions (e.g., weight portions **240** and **250**). In one example, weight portions **240** and **250** may have the same mass and coupled the toe-side weight port **220** and the heel-side weight port **230**, respectively, to provide balanced weighting to the golf club head **100**. In another example,

weight portion **240** may have a greater mass than weight portion **250**, and weight portions **240** and **250** may be coupled the toe-side weight port **220** and the heel-side weight port **230**, respectively, to counteract an individual's tendency to pull putts by restricting club head rotation and face closure during a putting stroke. In yet another example, weight portion **240** may have a lower mass than weight portion **250**, and weight portions **240** and **250** may be coupled the toe-side weight port **220** and the heel-side weight port **230**, respectively, to counteract an individual's tendency to push putts by promoting club head rotation and face closure during a putting stroke. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

The mass properties of the golf club head **100** may be further adjusted with an interchangeable hosel system that may utilize a hosel bore instead of a fixed hosel (e.g., an integral portion of the body portion **110**). Accordingly, the hosel bore **135** may be configured to receive a variety of hosel types to adjust the balance of the golf club head **100**. In one example, as shown in FIG. **13**, a hosel portion **1300** configured as a heel-shafted hosel type may be coupled to the hosel bore **135** to orient a shaft axis **1310** away from a CG location **1320** and more toward a heel portion **1330** of the golf club head **100** to increase the inertial force required to rotate the golf club head **100** during a putting stroke. A heel-shafted hosel may produce a large degree of toe hang (e.g., greater than 45 degrees) and may be preferred by individuals that tend to over-rotate the golf club head **100**, thereby producing a closed face at impact. In another example, as shown in FIG. **14**, a hosel portion **1400** configured as a double bend hosel type may be coupled to the hosel bore **135** to orient a shaft axis **1410** through a CG location **1420** to moderately reduce the inertial force required to rotate the golf club head **100** during a putting stroke. Alternatively, an armlock hosel type (not shown) may be used to accomplish the same. A double bend hosel type or armlock hosel type may promote face balancing and may be preferred by individuals that tend to under-rotate the golf club head **100** thereby producing an open face at impact. In yet another example, as shown in FIG. **15**, a hosel portion **1500** configured as a plumber's neck hosel type may be coupled to the golf club head **100** to orient a shaft axis **1510** away from a heel portion **1520** and closer to a CG location **1530** of the golf club head **100** to significantly reduce the inertial force required to rotate the golf club head **100** during a putting stroke. A plumber's neck hosel type may produce a moderate degree of toe hang (e.g., less than or equal to 45 degrees) and may be preferred by individuals exhibiting mild levels of club head rotation. By employing one or both of the weighting system and the interchangeable hosel system described herein, the mass properties of the golf club head **100** may be adjusted to fit the needs of any individual regardless of putting style or swing tendencies while providing greater forgiveness due to the golf club head **100** benefitting from higher MOI compared to conventional solid-body golf club heads. While the examples and figures may describe and depict a hosel bore and a separate hosel portion, the apparatus, methods, and articles of manufacture described herein may include a hosel portion integral to the body portion. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

In the example of FIGS. **16-30**, a golf club head **1600** is generally shown as a mid-mallet or mallet type putter. The golf club head **1600** may include a body portion **1610** having a toe portion **1620**, a heel portion **1630**, a hosel bore **1635**, a front portion **1640**, a face portion **1645**, a rear portion

1650, a top portion **1660**, and a sole portion **1710**. The body portion **1610** may be made from any of the materials described herein or described in any of the incorporated by reference patent documents. The face portion **1645** may be located at the front portion **1640** and may define a front surface of the body portion **1610** for striking a golf ball (e.g., shown as golf ball **2900**). The face portion **1645** may be generally planar and may have a loft angle that is greater than or equal to two degrees and less than or equal to four degrees. The sole portion **1710** may include a plurality of weight ports (e.g., shown as a toe-side weight port **1711** and a heel-side weight port **1712**) each configured to receive a complementary weight portion (e.g., shown as weight portions **1721** and **1722**). Weight portions **1721** and **1722** may be configured as interchangeable screws threadedly coupled to the toe-side weight port **1711** and the heel-side weight port **1712**, respectively. Weight portions **1721** and **1722** may have similar or different material compositions and/or physical properties (e.g., density, color, texture) with respect to each other and the body portion **1610**. The configuration and/or properties of each of the weight ports and the weight portions may be similar in many respects to any of the weight ports and weight portions described herein or described in any of the incorporated by reference patent documents. The hosel bore **1635** may be located at or proximate the heel portion **1630** and may be configured to receive a hosel portion **1670**. The hosel portion **1670** may have similar or different material compositions and/or physical properties with respect to the body portion **1610**. The hosel portion **1670** may be made from any of the materials described herein or described in any of the incorporated by reference patent documents. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

In one example, the body portion **1610** may be defined by a forward portion **1611** and an aft portion **1612** extending rearward from the forward portion **1611**. The forward portion **1611** may extend from the face portion **1645** to a back wall portion **1730** opposite to the face portion **1645**. The back wall portion **1730** may extend between the toe portion **1620** and the heel portion **1630** and may be located between the front portion **1640** and the rear portion **1650**. The aft portion **1612** may extend rearward from the back wall portion **1730** and may include the rear portion **1650**. In one example, the aft portion **1612** may be cantilevered from the forward portion **1611**. Collectively, the forward portion **1611** and the aft portion **1612** may partially enclose a central void space **1613** of the body portion **1610**. In one example, the central void space **1613** may extend in a top-to-sole direction. In another example, the central void space **1613** may extend in a toe-to-heel direction. The body portion **1610** may include an interior cavity **2500** that interfaces with an opening **2510** located at the sole portion **1710**. The interior cavity **2500** may be provided within the forward portion **1611** and may extend between the face portion **1645** and the back wall portion **1730**. The interior cavity **2500** may also extend between the top portion **1660** and the sole portion **1710** and between the toe portion **1620** and the heel portion **1630**. A sole plate **1740** may be coupled to the body portion **1610** to close the opening **2510**. The sole plate **1740** may have similar or different material compositions and/or physical properties as the body portion **1610**. The sole plate **1740** may include any of the materials described herein or described in any of the incorporated by reference patent documents. In one example, the sole plate **1740** may be made from a material having a density that is less than a density of the body portion **1610** to enable greater discre-

tionary mass placement in other regions of the golf club head **1600**. The sole plate **1740** may be welded to the body portion **1610** or otherwise attached to the body portion **1610** using adhesive and/or mechanical fasteners. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

A central portion **1680** may be coupled to the body portion **1610** and may be configured to extend across the central void space **1613** in a front-to-rear direction. The central portion **1680** may be located at the top portion **1660** and may bridge the front portion **1640** and the rear portion **1650**. In one example, the central portion **1680** may extend farther rearward than the rearmost extent of the aft portion **1612**. The central portion **1680** may include any type of alignment aid. In one example, as illustrated in FIGS. **16-30**, an alignment aid **1685** is generally shown as a line extending longitudinally across the central portion **1680** to assist an individual with addressing the golf club head **100** to the golf ball **2900**. The central portion **1680** may have similar or different material compositions and/or physical properties as the body portion **1610**. The central portion **1680** may include any of the materials described herein or described in any of the incorporated by reference patent documents. In one example, the central portion **1680** may be made from a material having a density that is less than a density of the body portion **1610** to enable greater discretionary mass placement in other regions of the golf club head **1600**. Accordingly, the central portion **1680** may be provided as a separate piece attached to the body portion **1610**. In another example, the central portion **1680** may be co-manufactured with the body portion **1610** so as to be a continuous one-piece part with the body portion **1610**. In another example, all or portions of the central portion **1680** may have different surface visual effects such as surface texture, geometric shapes (e.g., a circle resembling a golf ball), alphanumeric information and/or one or more colors to further assist in the alignment aid functionality of the central portion **1680**. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

A lower portion **1690** may be coupled to the body portion **1610** and may provide structural support to the aft portion **1612**. The lower portion **1690** may undergird the aft portion **1612** and may be configured with a complementary shape to the aft portion **1612** so as not to obstruct the central void space **1613**. The lower portion **1690** may have similar or different material compositions and/or physical properties as the body portion **1610**. The lower portion **1690** may include any of the materials described herein or described in any of the incorporated by reference patent documents. In one example, the lower portion **1690** may be made from a material having a density that is less than a density of the body portion **1610** to enable greater discretionary mass placement in other regions of the golf club head **1600**. To this end, the lower portion **1690** may be made from a lightweight material that is resistant to deformation (e.g., a composite material). Accordingly, the lower portion **1690** may be provided as a separate piece attached to the body portion **1610**. In another example, the lower portion **1690** may be co-manufactured with the body portion **1610** so as to be a continuous one-piece part with the body portion **1610**. In one example, the lower portion **1690** may cooperate with the back wall portion **1730** to frictionally engage a golf ball. Accordingly, the golf club head **1600** may be used to pick up a golf ball by positioning the central void space **1613** over the golf ball and pressing the golf club head **1600** onto the golf ball such that the golf ball becomes lodged or pinched between the lower portion **1690** and the back wall

portion **1730**. In one example, the ball engaging surfaces of the lower portion **1690** and the back wall portion **1730** may be textured to improve frictional engagement with the golf ball. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

In assembly, the central portion **1680** may include a front end portion **2411** and a rear end portion **2412**, each of which may be at least partially received in a complementary recess (e.g., shown as recess **2421** and recess **2422**, respectively) at the top of the forward portion **1611** and at the top and rear of the aft portion **1612**, respectively. The front end portion **2411** may include a boss **2520** received in a complementary indent **2430** in recess **2421** to properly position the central portion **1680** atop the body portion **1610**. The boss **2520** may include a fastener port **2521** aligned with an opening **2431** extending through indent **2430** and interfacing with the interior cavity **2500**. The fastener port **2521** and opening **2431** may be aligned with a through-port **2440** connected to a depression **2530** in an outer surface **2540** of the sole plate **1740**. A mechanical fastener **2451** may be inserted into the through-port **2440** and threadedly engaged to fastener port **2521** via opening **2431** thereby joining the front end portion **2411** to the body portion **1610** and the sole plate **1740**. The rear end portion **2412** may include a fastener port **2522** aligned with an opening **2432** in recess **2422** and a through-port **2461** connected to a depression **2550** in a lower outer surface **2560** of the lower portion **1690**. A mechanical fastener **2452** may be inserted into the through-port **2461** and threadedly engaged to fastener port **2522** via opening **2432** thereby joining the rear end portion **2412** to the body portion **1610** and the lower portion **1690**. In addition to mechanical fastener **2452**, the lower portion **1690** may be attached to the underside of the aft portion **1612** using an adhesive. A complementary cover portion (e.g., shown as cover portions **1751** and **1752**) may be received in each of depressions **2530** and **2550** to conceal corresponding mechanical fasteners **2451** and **2452**, respectively. In one embodiment, cover portions **1751** and **1752** may each be configured as decorative badges and may be attached to corresponding depressions **2530** and **2550** using adhesive. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

The interior cavity **2500** may be partially or entirely filled with a filler material **2470**. The filler material **2470** may include any of the filler materials described herein or described in any of the incorporated by reference patent documents. In one example, the filler material **2470** may include a low-density polymer material. In another example, the filler material **2470** may be a solid piece manufactured in the shape of the interior cavity **2500** so as to be insertable into the interior cavity **2500** through the opening **2510**. In yet another example, the filler material **2470** may be an injectable liquid or fluid material provided to the interior cavity **2500** via one or more openings (e.g., opening **2510**) and later hardened at ambient conditions or during a curing process. The filler material **2470** may be disposed within the interior cavity **2500** to contact and reinforce a rear surface **2570** of the face portion **1645** to reduce face deformation and also dampen sound and vibration. Additionally, the filler material **2470** may at least partially encase the depression **2530** and the through-port **2440** of the sole plate **1740**. The filler material **2470** may provide structural support to the cavity walls and/or the face portion **1645** of the golf club head **1600**. Accordingly, the cavity walls may be relatively thin to accommodate a greater amount of the filler material **2470** in the interior cavity **2500** and to provide increased discretionary mass, and the face portion **1645** may be

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relatively thin to provide a greater flexibility or rebounding effect. In one example, one or more cavity walls may have a thickness (constant or average) greater than or equal to 0.035 inch (0.889 mm) and less than or equal to 0.060 inch (1.524 mm). In another example, the face portion **1645** may have a face thickness **2600** in one or more areas greater than or equal to 0.035 inch (0.889 mm) and less than or equal to 0.055 inch (1.397 mm) (e.g., first face thickness **1240**) and greater than or equal to 0.035 inch (0.889 mm) and less than or equal to 0.045 inch (1.143 mm) (e.g., second face thickness **1260**) as described herein with reference to FIG. **12**. In another example, the face portion **1645** may have a face thickness **2600** in one or more areas less than or equal to 0.035 inch (0.889 mm) or greater than or equal to 0.055 inch (1.397 mm). The face portion **1645** including the face thickness at one or more area on the face portion **1645** may be similar in many respects to any face portion (e.g., face portion **145**) described herein or described in any of the incorporated by reference patent documents. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

With reference to FIGS. **26** and **27**, In one example, the golf club head **1600** may have at least one cross-section where the filler material **2470** has a length **2610** in a front-to-rear direction greater than or equal to 80% of a length **2620** between the face portion **1645** and the back wall portion **1730** in the front-to-rear direction. In another example, the golf club head **1600** may have at least one cross-section where the filler material **2470** has a length **2610** in a front-to-rear direction greater than or equal to 65% of a length **2620** between the face portion **1645** and the back wall portion **1730** in the front-to-rear direction. With reference to FIG. **26**, in one example, the golf club head **1600** may have at least one cross-section where the filler material **2470** may have a length **2630** in a top-to-sole direction greater than or equal to 70% of a length **2640** of the body portion **1610** in the top-to-sole direction. In another example, the golf club head **1600** may have at least one cross-section where the filler material **2470** may have a length **2630** in a top-to-sole direction greater than or equal to 55% of a length **2640** of the body portion **1610** in the top-to-sole direction. With reference to FIG. **27**, in one example, the golf club head **1600** may have at least one cross-section wherein the filler material **2470** has a length **2710** in a toe-to-heel direction greater than or equal to 90% of a length **2720** of the body portion **1610** in the toe-to-heel direction. In another example, the golf club head **1600** may have at least one cross-section wherein the filler material **2470** has a length **2710** in a toe-to-heel direction greater than or equal to 75% of a length **2720** of the body portion **1610** in the toe-to-heel direction. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

The filler material **2470** may have a density less than a density of the body portion **1610** to allow for more discretionary mass to be allocated to other areas of the golf club head **1600**. With respect to the present example, the filler material **2470** may have a fill volume (V_f) greater than or equal to 30% and less than or equal to 40% of a total volume (V_t) of the golf club head **1600** (e.g., with or without the hosel portion **1670**) and a fill mass (M_f) greater than or equal to 5% and less than or equal to 10% of a total mass (M_t) of the golf club head **1600** (e.g., with or without the hosel portion **1670**). In one example, the total volume (V_t) of the golf club head **1600** may be greater than or equal to 5.40 inch³ (88.490 cm³) and less than or equal to 5.90 inch³ (96.684 cm³) and the total mass (M_t) of the golf club head

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1600 may be greater than or equal to 320 grams and less than or equal to 370 grams. As described herein, selecting the filler material **2470** from a lightweight or low-density material and increasing the fill volume (V_f) of the filler material **2470** relative to the total volume (V_t) of the golf club head **1600** may enable greater discretionary mass placement toward a periphery of the golf club head **1600** to optimize certain mass properties of the golf club head **1600** such as MOI, CG location, and mass balance while maintaining a suitable club head weight. In one example, a ratio between a percentage fill volume

$$\left(\frac{V_f}{V_t} \times 100\right)$$

and a percentage fill mass

$$\left(\frac{M_f}{M_t} \times 100\right)$$

of the filler material **2470** may be greater than or equal to 3.00 and the golf club head **1600** may have a MOI greater than or equal to 4000 g-cm². In another example, the golf club head **1600** may have a MOI greater than or equal to 4250 g-cm². In yet another example, the golf club head **1600** may have a MOI greater than or equal to 4500 g-cm². The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

As described herein, the golf club head **1600** may include a weighting system that may utilize a plurality of weight ports (e.g., toe-side weight port **1711** and heel-side weight port **1712**) and corresponding interchangeable weight portions (e.g., weight portions **1721** and **1722**) to adjust the mass properties of the golf club head **1600**. Additionally, as described herein, the hosel bore **1635** may be configured to receive any of the hosel types described herein or described in any of the incorporated by reference patent documents to further adjust the mass properties of the golf club head **1600**. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

With reference to FIG. **28**, the hosel bore **1635** is shown coupled to the hosel portion **1670**. The hosel portion **1670** may have a serpentine shape and may include a first bend portion **2810**, a second bend portion **2820**, and a tip portion **2830** configured to be received inside a golf shaft **2800**. In one example, as shown in FIG. **20**, the hosel bore **1635** and the hosel portion **1670** may be located at or proximate the heel portion **1630** and may be entirely disposed within a front space **2001** bounded by a center longitudinal plane **2010**, a heel boundary plane **2020** parallel to the center longitudinal plane **2010**, a front boundary plane **2030**, and a first lateral plane **2040** parallel to the front boundary plane **2030** and separated from the front boundary plane **2030** by a distance **2050** equal to one third of a total length **2060** of the golf club head **1600** in the front-to-rear direction. The hosel bore **1635** and the hosel portion **1670** may be intersected by a heel dividing plane **2070** parallel to and equidistant from the center longitudinal plane **2010** and the heel boundary plane **2020**. In another example, the hosel bore **1635** and the hosel portion **1670** may be partially or entirely disposed within a middle space **2002** bounded by the center longitudinal plane **2010**, the heel boundary plane **2020**, the first lateral plane **2040**, and a second lateral plane **2080** parallel to the front boundary plane **2030** and separated from

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the front boundary plane **2030** by a distance **2085** equal to two thirds the total length **2060** of the golf club head **1600** in the front-to-rear direction. In yet another example, the hosel bore **1635** and the hosel portion **1670** may be partially or entirely disposed within a rear space **2003** bounded by the center longitudinal plane **2010**, the heel boundary plane **2020**, the second lateral plane **2080** and a third lateral plane **2090** parallel to the front boundary plane **2030** and separated from the front boundary plane **2030** by a distance **2095** equal to the total length **2060** of the golf club head **1600** in the front-to-rear direction. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

Referring still to FIG. **28**, the hosel portion **1670** may be configured to orient a shaft axis **2870** toe-ward relative to a CG location **2880** of the golf club head **1600**. In other words, a distance between the shaft axis **2870** and the center longitudinal plane **2010** may increase in a heel-to-toe direction as the shaft axis **2870** extends from a location of intersection of the shaft axis **2870** and the center longitudinal plane **2010** toward the sole portion **1710**. The shaft axis **2870** may intersect with the center longitudinal plane **2010** at or proximate (e.g., below) the alignment aid **1685** such that the hosel portion **1670** provides an unobstructed view **2910** of the alignment aid **1685** when the golf club head **1600** is addressed to the golf ball **2900** as shown in FIG. **29**. In one example, a distance **2890** between the shaft axis **2870** and the CG location **2880** in the heel-to-toe direction may be greater than or equal to 3 mm and less than or equal to 10 mm toward the toe portion **1620**. In another example, the distance **2890** between the shaft axis **2870** and the CG location **2880** may be greater than or equal to 4 mm and less than or equal to 8 mm toward the toe portion **1620**. In yet another example, the distance **2890** between the shaft axis **2870** and the CG location **2880** may be greater than or equal to 5 mm and less than or equal to 6 mm toward the toe portion **1620**. With respect to each of the foregoing examples, the shaft axis **2870** may have zero offset with the CG location **2880** in the front-to-rear direction or may be offset from the CG location **2880** toward the front portion **1640** or the rear portion **1650**. In one example, the shaft axis **2870** and the CG location **2880** may be offset by a distance greater than or equal to 0 mm and less than or equal to 5 mm in the front-to-rear direction toward the front portion **1640** or the rear portion **1650**. In another example, the shaft axis **2870** and the CG location **2880** may be offset by a distance greater than or equal to 0 mm and less than or equal to 4 mm in the front-to-rear direction toward the front portion **1640** or the rear portion **1650**. In yet another example, the shaft axis **2870** and the CG location **2880** may be offset by a distance greater than or equal to 0 mm and less than or equal to 3 mm in the front-to-rear direction toward the front portion **1640** or the rear portion **1650**. Additionally, the shaft axis **2870** may have zero offset with the CG location **2880** in the top-to-sole direction or may be offset from the CG location **2880** toward the top portion **1660** or the sole portion **1710**. In one example, the shaft axis **2870** and the CG location **2880** may be offset by a distance greater than or equal to 0 mm and less than or equal to 5 mm in the top-to-sole direction toward the top portion **1660** or the sole portion **1710**. In another example, the shaft axis **2870** and the CG location **2880** may be offset by a distance greater than or equal to 0 mm and less than or equal to 4 mm in the top-to-sole direction toward the top portion **1660** or the sole portion **1710**. In yet another example, the shaft axis **2870** and CG location **2880** may be offset by a distance greater than or equal to 0 mm and less than or equal to 3 mm in the top-to-sole direction toward the

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top portion **1660** or the sole portion **1710**. The amount of offset, if any, between the shaft axis **2870** and the CG location **2880** in the front-to-rear direction and/or the top-to-sole direction may be similar or different to the amount of offset between the shaft axis **2870** and the CG location **2880** in the heel-to-toe direction. Accordingly, any distance separating the shaft axis **2870** from the CG location **2880** in the front-to-rear direction and/or the top-to-sole direction may be similar or different to any of the distance **2890** examples described herein. In examples where the shaft axis **2870** is offset from the CG location **2880** toward the toe portion **1620**, the hosel portion **1670** may allow the golf club head **1600** to balance in a toe-up position as shown in FIG. **30**. For example, when the hosel portion **1670** is connected to a golf shaft resting on a flat horizontal surface and the golf club head **1600** is allowed to hang freely over an edge, the toe portion **1620** of the golf club head **1600** may point naturally in an upward direction such that a hang angle **3000** may be formed between a face plane **3010** and a horizontal plane **3020** parallel to or coinciding with the flat horizontal surface. In one example, the hang angle **3000** may be greater than 0 degrees and less than or equal to 45 degrees. In another example, the hang angle **3000** may be greater than or equal to 45 degrees and less than or equal to 90 degrees. In yet another example, the hang angle **3000** may be greater than or equal to 90 degrees and less than or equal to 135 degrees. In yet another example, the hang angle **3000** may be greater than or equal to 135 degrees and less than 180 degrees. A golf club employing a golf club head that balances toe-up may provide a similar feel to a face balanced golf club while also promoting greater face awareness compared to a conventional face balanced golf club. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

FIG. **31** depicts one manner by which any of the golf club heads described herein may be manufactured. In the example of FIG. **31**, a process **3100** may begin with providing a body portion having a hosel bore and an interior cavity that interfaces with an opening (block **3110**). A filler material may be provided into the interior cavity via the opening (block **3120**). In one example, the filler material may be adapted as an injectable liquid polymer material and later hardened using a curing process. In another example, the filler material may be adapted as an insertable solid polymer material. As described herein, the filler material may reinforce a face portion of the golf club head and may enable greater discretionary mass placement in other areas of the golf club head to optimize certain mass properties such as MOI, CG location, and mass balance. A cover portion may be provided to close the opening and enclose the interior cavity (block **3130**). A plurality of weight portions may be coupled to the body portion to adjust certain mass properties of the golf club head (block **3140**). A hosel portion may be coupled to the hosel bore to further adjust the mass properties of the golf club head (**3150**). As described herein, the hosel portion may be selected from a variety of different hosel types based on a golfer's swing type and swing tendencies. In one example, the hosel portion may be configured to orient a shaft axis toe-ward relative to a CG location of the golf club head such that the golf club head balances in a toe-up position. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

In the example of FIG. **32**, a golf club **3200** may include a golf club head **3210**, a shaft **3220**, a grip **3230**, and a hosel portion **3240**. The shaft **3220** may have a tip end portion **3221** and a butt end portion **3222**. The butt end portion **3222**

of the shaft **3220** may be coupled to the grip **3230** and the tip end portion **3221** of the shaft **3220** may be coupled to the hosel portion **3240**. The hosel portion **3240** may be coupled to the golf club head **3210** via a hosel bore as described herein. The golf club head **3210** and the hosel portion **3240** may include any of the golf club heads and hosel types described herein. The shaft **3220** may be formed from a metal material, a composite material, or any other suitable material or combination of materials. The grip **3230** may be formed from a rubber material, a polymer material, or any other suitable material or combination of materials. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

In the example of FIGS. 33-35, a grip attachment device **3300** is shown for demonstrating how a putter-type golf club behaves during a putting stroke. As described below, the grip attachment device **3300** allows the golf club to swing freely while highlighting the effects of torque and natural rotation. In the illustrated example, the grip attachment device **3300** may be configured to encase a grip **3310** and an upper portion of a shaft **3320** of a golf club and may be configured to enable the golf club to freely rotate about a shaft axis **3325**. When the grip attachment device is fitted to a golf club and the golf club is swung while holding the grip attachment device **3300**, the golf club may undergo rotation (e.g., twist) based on certain mass balance properties of the golf club head. For example, when a putting stroke is performed using the grip attachment device **3300** fitted to a golf club employing the golf club head **1600** described herein, the golf club head **1600** may remain square or substantially square to the stroke path, or in other words, may exhibit little to no twist despite the golf club being free to rotate. This may be attributed to the golf club head **1600** having a hosel portion (e.g., hosel portion **1670**) located at a heel portion (e.g., heel portion **1630**) and configured to direct a shaft axis (e.g., shaft axis **2870**) toe-ward from a center of gravity location (e.g., center of gravity location **2880**). Accordingly, a golf club utilizing the golf club head **1600** may be considered torque resistant as a result of exhibiting little to no rotation when a putting stroke is performed using the grip attachment device **3300**. In contrast, a golf club that is not torque resistant may exhibit moderate to high degrees of rotation when a putting stroke is performed using the grip attachment device **3300**. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

In the example of FIGS. 33-35, the grip attachment device **3300** may include a housing generally shown as a tubular portion **3330** having an upper end portion **3331** and a lower end portion **3332**. The tubular portion **3330** may be made from a transparent or semitransparent polymer material to enable a user to see the contents contained therein. A plug portion **3340** may be coupled to the upper end portion **3331** and a clamp portion **3350** may be coupled to the lower end portion **3332**. In one example, the plug portion **3340** may be removably coupled (e.g., frictionally coupled) to the upper end portion **3331**. In another example, the plug portion **3340** may be fixedly secured to the upper end portion **3331** using mechanical fasteners, adhesive, and the like. In yet another example, the plug portion **3340** may be integral with the tubular portion. The plug portion **3340** may include a port **3410** with an opening **3415** in communication with an interior space of the tubular portion **3330**. A torque assembly **3420** may be disposed within the port **3410** and may include a spacer **3430** interposed between a plurality of washers (e.g., shown as washers **3441**, **3442**, **3443**, and **3444**) and a plurality of rolling bearings (e.g., shown as rolling bearings **3451** and **3452**). In the illustrated example, the spacer **3430**

may include a central body portion **3435** that may extend through the plurality of washers and the plurality of rolling bearings. The central body portion **3435** may protrude through the opening **3415** and into the interior space of the tubular portion **3330**. The spacer **3430**, the plurality of washers, and the plurality of rolling bearings may be concentrically aligned about a center longitudinal axis **3460** of the tubular portion **3330**. The torque assembly **3420** may be secured inside the port **3410** via a retaining ring **3470** coupled to the port **3410** and configured to prevent the torque assembly **3420** from becoming dislodged from the port **3410**. A fastener **3480** may extend downward through the central body portion **3435** and into the interior space of the tubular portion **3330**. In one example, the fastener **3480** may be configured as a screw threaded through the central body portion **3435**. The spacer **3430** may be made of a low friction material such as, but not limited to, Delrin®, to enable the spacer **3430** to freely rotate about the center longitudinal axis **3460** of the tubular portion **3330**. In the illustrated example, the clamp portion **3350** may be removably coupled to the lower end portion **3332**. The clamp portion **3350** may have a clamshell design and may be spring-loaded to an open position when the clamp portion **3350** is removed from the tubular portion **3330**. The clamp portion **3350** may have one or more detents (e.g., detent **3360** in FIG. 33), each configured to engage a corresponding aperture (e.g., aperture **3370** in FIG. 33) at the lower end portion **3332** of the tubular portion **3330**. The clamp portion **3350** may additionally include a bore **3510** for encasing an upper portion of the shaft **3320**. The bore **3510** may be concentrically aligned with the center longitudinal axis **3460** of the tubular portion **3330**. Additionally, the bore **3510** may be at least partially lined with a low friction material **3520** such as, but not limited to, a wool lining. In this manner, the shaft **3320** may freely rotate within the clamp portion **3350** with little impediment. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

In assembly, the clamp portion **3350** may be removed from the tubular portion **3330** to enable the grip **3310** and shaft **3320** portion to be received inside the tubular portion **3330** via the lower end portion **3332**. The fastener **3480** may be screwed into the top of the grip **3310**, thereby securing the grip **3310** to the spacer **3430** and allowing the golf club and the spacer **3430** to freely rotate in unison. In one example, the fastener **3480** may be screwed into a grip hole commonly found at the top of conventional grips. In another example, a golf club may be outfitted with a dedicated grip configured to readily accept the fastener **3480**. Once the grip is properly secured within the tubular portion **3330**, the clamp portion **3350** may be returned to a closed position to encase the shaft **3320** within the bore **3510** and subsequently inserted back into the tubular portion **3330** via the lower end portion **3332** by sliding the clamp portion **3350** upward along the shaft **3320** until the detent **3360** engages the corresponding aperture **3370** at the lower end portion **3332** of the tubular portion **3330**. In practice, the grip attachment device **3300** may be used by fitters, instructors, and/or golf equipment retailers and manufacturers to assist golfers in understanding the effects of torque and rotation for a particular golf club. By providing visual confirmation of a golf club's natural tendency to rotate or twist during a putting stroke, golfers may be better informed during the golf club selection process and may be able to quickly identify a golf club that best suits their individual skillsets and tendencies. For example, a golf club employing the golf club head **1600** described herein may be desirable for golfers seeking to

simplify their putting strokes. Since a golf club employing the golf club head **1600** may exhibit a greater level of torque resistance, a golfer may experience little to no need to manipulate the golf club to achieve a square face at impact, potentially leading to more consistent and accurate putts. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

In the example of FIGS. **36-41**, a golf club head **3600** is generally shown as a mid-mallet or mallet type putter head. However, the principles described herein may be similarly adopted to a blade or other type of putter head. As described in greater detail below, the golf club head **3600** may be similar in some respects to the golf club head **1600** described herein with reference to FIGS. **16-30**. In particular, the golf club head **3600** may be configured with mass properties that promote higher levels of torque resistance while maintaining stability throughout a putting stroke. Accordingly, a golf club featuring the golf club head **3600** of the present disclosure may exhibit minimal rotation or twisting when a putting stroke is performed using the grip attachment device **3300** described previously herein. As described herein, a golf club designed to minimize rotation or twisting during a putting stroke may reduce or eliminate club manipulations in order to achieve a square face at impact. As a result, the golf club may promote a more consistent stroke path and face angle at impact. Having a square face at impact may enable golfers to hit putts closer to the intended target line, thereby improving precision and inspiring confidence. Additionally, consistent face alignment and reduced twisting may lead to more predictable ball speed and roll, potentially improving distance control. The golf club may suit a variety of swing types, including both straight-back-straight-through putting strokes and arc putting strokes. Given these benefits, a golf club featuring the golf club head **3600** may offer a versatile option capable of satisfying the needs of different golfers, particularly those struggling to square the face. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

In the example of FIGS. **36-41**, the golf club head **3600** may include a body portion **3610** having a toe portion **3620** with a toe portion edge **3622**, a heel portion **3630** with a heel portion edge **3632**, a hosel portion **3635** coupled to a hosel bore **3636**, a front portion **3640** with a face portion **3645**, a rear portion **3650** with a rear portion edge **3652**, a top portion **3660** with a top portion edge **3662**, and a sole portion **3710** with a sole portion edge **3712**. An alignment aid **3670** may be located at the top portion **3660** and may extend between the front portion **3640** and the rear portion **3650**. The alignment aid **3670** may be configured as a recessed line that coincides with a center longitudinal plane **3720**, which may be a plane that bisects the body portion **3610** to define the toe portion **3620** and the heel portion **3630**. The alignment aid **3670** may be segmented into a first portion **3671** located at a forward portion **3680** of the body portion **3610** and a second portion **3672** located at a ledge portion **3690** positioned aft of the forward portion **3680** and extending between a toe-side wing portion **3621** and a heel-side wing portion **3631**. When the golf club head **3600** is viewed from above (e.g., at an address position to strike a golf ball such as the golf ball **2900**), the first portion **3671** of the alignment aid **3670** may align with the second portion **3672** of the alignment aid **3670** to give the appearance of a single straight line. The toe-side wing portion **3621** and the heel-side wing portion **3631** may each extend rearwardly from the forward portion **3680** and may cooperate with the ledge portion **3690** to define a gap **3695**. In one example, the golf club head **3600** may be used to pick up a golf ball by

positioning the gap **3695** over the golf ball and pressing the golf club head **3600** onto the golf ball such that the golf ball becomes lodged or pinched between the toe-side wing portion **3621** and the heel-side wing portion **3631**. In the illustrated example, the hosel bore **3636** may be formed in a hosel receptacle portion **3685** extending rearwardly from the forward portion **3680** and extending upwardly from the heel-side wing portion **3631**. In one example, the hosel receptacle portion **3685** may be integral with the heel-side wing portion **3631** and a back wall portion **3681** of the forward portion **3680** that may generally oppose the face portion **3645**. In another example, the hosel receptacle portion **3685** may be separate from the forward portion **3680** and may correspond to a standalone structure extending upwardly from the heel-side wing portion **3631**. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

The body portion **3610** may be made from any of the material and/or processes described herein or described in any of the incorporated by reference documents. Additionally, the body portion **3610** may include any of the features described herein or described in any of the incorporated by reference documents. Such features may include, but are not limited to, one or more weight ports and corresponding weight portions and/or a filler material housed within a cavity of the body portion **3610**. For purposes of clarity, the face portion **3645** is generally shown having a flat surface without any face features (e.g., grooves, protrusions, etc.). However, in practice, the face portion **3645** may be configured according to any example face portion described herein or described in the incorporated by reference documents. In one example, the face portion may be configured with a face pattern similar to the face pattern described herein with reference to FIGS. **10-12** and may be supported by a filler material located behind at least a portion of the face portion **3645**. Accordingly, the face portion **3645** may benefit from a thin face design and may be configured with any of the face thicknesses described herein or described in the incorporated by reference documents. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

FIGS. **37-39** show a reference coordinate system for the golf club head **3600**. The coordinate system may include an origin **3730**, an x-axis **3731**, a y-axis **3732**, and a z-axis **3801**. In the illustrated example, the origin **3730** may correspond to a midpoint of a leading edge **3740** of the golf club head **3600**. The leading edge **3740** may join the sole portion **3710** to the face portion **3645** and the midpoint of a leading edge **3740** may be located halfway between the toe portion edge **3622** and the heel portion edge **3632**. The x-axis **3731** may extend between the toe portion **3620** and the heel portion **3630**. The portion of the x-axis **3731** extending from the origin **3730** in a toe-ward direction or toward the toe portion edge **3622** may include negative x-axis values, and the portion of the x-axis **3731** extending from the origin **3730** in a heel-ward direction or toward the heel portion edge **3632** may include positive x-axis values. The y-axis **3732** may extend between the top portion **3660** and the sole portion **3710**. The portion of the y-axis **3732** extending from the origin **3730** in a sole-to-top direction may include positive y-axis values whereas negative y-axis values may fall below the sole portion **3710**. The z-axis **3801** may extend between the front portion **3640** and the rear portion **3650**. The portion of the z-axis **3801** extending from the origin **3730** in a front-to-rear direction or toward the rear portion edge **3652** may encompass negative z-axis value whereas positive z-axis values may typically fall forward of

the face portion **3645**. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

FIG. **39** shows a quadrant system about the golf club head **3600**. The quadrant system may include a center lateral plane **3910** that intersects with the center longitudinal plane **3720** to define a front-toe quadrant **3921**, a front-heel quadrant **3922**, a rear-toe quadrant **3923**, and a rear-heel quadrant **3924**. In one example, the center longitudinal plane **3720** may equally divide a length **3931** of the body portion **3610** between the toe portion **3620** and the heel portion **3630** whereas the center lateral plane **3910** may equally divide a length **3932** of the body portion **3610** between the front portion **3640** and the rear portion **3650**. The center longitudinal plane **3720** may coincide with the y-axis **3732** and the z-axis **3801** and may be perpendicular with a ground plane **3735** that may include any flat surface on which the golf club head **3600** rests at an address position to strike a golf ball. In one example the ground plane **3735** may coincide with the x-axis **3731** and the z-axis **3801**. The center lateral plane **3910** may be perpendicular to the center longitudinal plane **3720** and the ground plane **3735** and may be parallel with the x-axis **3731**. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

In the example of FIGS. **36-41**, the golf club head **3600** may be configured with a center of gravity (CG) **3750** located heel-ward (e.g., having a positive x-axis value) from the center longitudinal plane **3720**. Positioning the CG **3750** heel-ward may enhance stability and feedback. For example, the golf club head **3600** may provide amplified feedback for off-center hits to assist golfers in refining their stroke and impact consistency. In the illustrated example, the CG **3750** may be located on an intersection between a horizontal plane **3760** and a vertical plane **4010** that intersect the center longitudinal plane **3720**. The horizontal plane **3760** may be parallel to the ground plane **3735** and may be perpendicular to the center longitudinal plane **3720**. The vertical plane **4010** may be perpendicular to the horizontal plane **3760**, the center longitudinal plane **3720**, and the ground plane **3735**. In one example, the CG **3750** may be located in the front-heel quadrant **3922** within a CG zone **3755** that may include any x-axis value on the x-axis **3731** bounded by the center longitudinal plane **3720** and a boundary plane **3770**, which is a plane that is parallel to the center longitudinal plane **3720** and defines a toe-side boundary of the hosel portion **3635**. In another example, the CG **3750** may include an x-axis value located outside the CG zone **3755** (i.e., on or beyond the boundary plane **3770** in the heel-ward direction). In yet another example, the CG **3750** may coincide with the center longitudinal plane **3720** or may be located in the front-toe quadrant **3921** thereby encompassing any negative x-axis values. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

In the example of FIGS. **36-41**, the CG **3750** coordinates may vary with respect to the y-axis **3732** and the z-axis **3801**. In the illustrated example, the CG **3750** may have a y-axis value located closer to the top portion **3660** than to the sole portion **3710** and a z-axis value located closer to the front portion **3640** than to the rear portion **3650**. The benefits of a higher CG **3750** may include improved roll consistency, better energy transfer, enhanced feel, and stability on mishits. For example, a higher CG **3750** may be positioned closer to a golf ball's equator, thereby imparting more immediate topspin, which may promote a smoother roll. Impact closer to the CG **3750** may reduce energy loss, resulting in greater strike efficiency. A higher CG **3750** may

also produce a crisp, solid feel at impact due to the face portion **3645** interacting more directly with the golf ball's equator. A higher CG **3750** may further be balanced with perimeter weighting to maintain stability and impart forgiveness on off-center strikes. The benefits of a forward CG **3750** may include increased ball speed and distance control, reduced face rotation, and improved responsiveness. For example, a forward CG **3750** may transfer more energy directly to a golf ball, resulting in higher ball speeds for the same swing effort, thereby aiding golfers in achieving better distance control. A forward CG **3750** may also lower a moment of inertia (MOI) of the golf club head **3600** about a shaft axis **3791**, which may reduce face rotation or twisting during a putting stroke. A forward CG **3750** may further produce a responsive feel at impact, aiding golfers in making smoother, more controlled putting strokes. In other examples, the CG **3750** may have a y-axis value located closer to the sole portion **3710** than to the top portion **3660** and/or a z-axis value located closer to the rear portion **3650** than to the front portion **3640**. As described herein, the particular CG **3750** location may be dictated through perimeter weighting (e.g., via weight portions), material properties of the body portion, one or more interior cavities partially or entirely filled with a filler material, and/or material properties and attachment location of the hosel portion **3635**. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

In the example of FIGS. **36-41**, the hosel portion **3635** may be coupled to the body portion **3610** at the hosel receptacle portion **3685** via the hosel bore **3636**. The hosel portion **3635** may be made from a material that is similar to or different from a material of the body portion **3610**. In the illustrated example, the hosel portion **3635** may be located within the front-heel quadrant **3922**. In another example, the hosel portion **3635** may be located within the rear-heel quadrant **3924**. In yet another example, the hosel portion **3635** may be located at the ledge portion **3690**. With respect to any of the examples provided herein, the hosel portion **3635** may be provided separately from the body portion **3610**. Alternatively, the hosel portion **3635** may be integral with the body portion **3610** or otherwise attached to the body portion **3610**. As shown in FIGS. **37** and **38**, the hosel portion **3635** may include an insert portion **3781** (e.g., shown in phantom line), a first bend portion **3782**, a neck portion **3783**, a second bend portion **3784**, and a tip portion **3785**. The insert portion **3781** may be configured to be received inside the hosel bore **3636** and may be coupled to the hosel bore **3636** using an adhesive (e.g., an epoxy), using a mechanical fastener, by being press fit, and/or with other types of securing devices. Additional examples of insertable hosel portions may be found in the incorporated by reference documents. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

The first bend portion **3782** may extend from the insert portion **3781** and may be configured to direct the neck portion **3783** in a toe-ward direction. In one example, as shown in FIG. **37**, the first bend portion **3782** may form a toe-ward bend angle **3786** in an x-y plane to impart a toe-ward lean to the neck portion **3783**. The toe-ward bend angle **3786** may correspond to an angle formed between the toe-side extents of the insert portion **3781** and the neck portion **3783** when viewing the golf club head **3600** from the vantage point of FIG. **37**. In one example, the toe-ward bend angle **3786** may be greater than or equal to 140 degrees and less than or equal to 160 degrees. In another example, the toe-ward bend angle **3786** may be greater than or equal to 145 degrees and less than or equal to 155 degrees. Addi-

tionally, in another example, the first bend portion **3782** may be configured to direct the neck portion **3783** in a rearward direction. The first bend portion **3782** may form a rearward bend angle **3811** in a y-z plane to impart a rearward lean to the neck portion **3783**. The rearward bend angle **3811** may correspond to an angle formed between the rearward extents of the insert portion **3781** and the neck portion **3783** when viewing the golf club head **3600** from the vantage point of FIG. **38**. In one example, the rearward bend angle **3811** may be greater than or equal to 165 degrees and less than or equal to 180 degrees. In another example, the rearward bend angle **3811** may be greater than or equal to 170 degrees and less than or equal to 175 degrees. The neck portion **3783** may extend linearly from the first bend portion **3782** in the toe-ward direction and the rearward direction and may terminate at the second bend portion **3784**. The second bend portion **3784** may extend from the neck portion **3783** and may be configured to direct the tip portion **3785** in a heel-ward direction. In one example, the second bend portion **3784** may have a heel-ward bend angle **3787** in an x-y plane to impart a heel-ward lean to the tip portion **3785**. The heel-ward bend angle **3787** may correspond to an angle formed between the heel-ward extents of the neck portion **3783** and the tip portion **3785** when viewing the golf club head **3600** from the vantage point of FIG. **37**. In one example, the heel-ward bend angle **3787** may be greater than or equal to 125 degrees and less than or equal to 140 degrees. In another example, the heel-ward bend angle **3787** may be greater than or equal to 130 degrees and less than or equal to 135 degrees. Additionally, in another example, the second bend portion **3784** may be configured to direct the tip portion **3785** in a frontward direction. The second bend portion **3784** may have a frontward bend angle **3812** in a y-z plane to impart a forward lean to the tip portion **3785**. The frontward bend angle **3812** may correspond to an angle formed between the frontward extents of the neck portion **3783** and the tip portion **3785** when viewing the golf club head **3600** from the vantage point of FIG. **38**. In one example, the frontward bend angle **3812** may be greater than or equal to 163 degrees and less than or equal to 178 degrees. In another example, the frontward bend angle **3812** may be greater than or equal to 168 degrees and less than or equal to 173 degrees. In other examples, the hosel portion **3635** may be configured to impart zero shaft lean or rearward shaft lean. In the illustrated example, the tip portion **3785** may be cylindrical or substantially cylindrical and may be coaxial with a shaft axis **3791** of the shaft **3790**. The shaft axis **3791** and the ground plane **3735** may form a lie angle **3795** measured when the golf club head **3600** rests flat (i.e., soled) on the ground plane **3735**. In the illustrated example, the lie angle **3795** of the golf club head **3600** may be 70 degrees or approximately 70 degrees. Alternatively, the lie angle **3795** may be greater than or less than 70 degrees. In one example, the lie angle **3795** may be in a range greater than or equal to 60 degrees and less than or equal to 80 degrees (e.g., the United States Golf Association (USGA) currently limits a lie angle of a conforming putter to 80 degrees). While the hosel portion **3635** is generally described herein as being internally coupled to the shaft **3790** via the tip portion **3785**, the hosel portion **3635** may be externally coupled to the shaft **3790** via a socket or otherwise coupled to the shaft **3790**. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

Referring to FIG. **37**, the golf club head **3600** may have a visibility angle **3796** in an x-y plane formed between the center longitudinal plane **3720** and a visibility plane **3797** that borders the toe-most extent (e.g., the second bend

portion **3784**) of the hosel portion **3635** and intersects the center longitudinal plane **3720** through the first portion **3671** of the alignment aid **3670**. The visibility angle **3796** may represent a range of sightlines that enable an individual to fully see the alignment aid **3670** at an address position to strike a golf ball. Accordingly, so long as an individual's sightline falls within the visibility angle **3796**, the individual may have full view of the alignment aid **3670**. As illustrated in the present example, the visibility angle **3796** may help position an individual's head over the golf club head **3600** at address such that the individual's sightline may be closer to vertical (i.e., 90 degrees). In this way, when an individual's head is positioned over the golf club head **3600** at address, the individual may have a clear line of sight to the intended target, which may improve aim, stability, and/or control. In the present example, the golf club head **3600** may have a visibility angle **3796** greater than or equal to 10 degrees and less than or equal to 16 degrees for a lie angle **3795** of 70 degrees. In another example, the visibility angle **3796** may be greater than or equal to 5 degrees and less than or equal to 20 degrees, irrespective of the lie angle **3795**. In another example, the golf club head **3600** may have a visibility angle greater than or equal to 0.5 degrees and less than or equal to 2 degrees for a lie angle greater than or equal to 75 degrees and less than or equal to 80 degrees). By providing a relatively narrow range of sightlines, a more optimal head position may be achieved to aid alignment at address. As an added visual feature, the hosel portion **3635**, configured in accordance with the teachings provided herein, may be concealed or substantially concealed by the shaft **3790** when viewing the golf club head **3600** at address, as exemplarily shown in FIG. **41**. In this manner, the golf club head **3600** may have a more minimalist appearance, giving the impression that the shaft **3790** is directly connected to the body portion **3610**. By reducing visual clutter, a golfer may better focus on alignment and setting up to make a putting a stroke. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

With reference to FIGS. **37** and **40**, the hosel portion **3635** may be configured to direct the shaft axis **3791** toe-ward from the CG **3750**. Directing the shaft axis **3791** toe-ward from the CG **3750** which is positioned heel-ward may impart certain performance properties that may enhance putting performance for many golfers. The synergistic effects of positioning the CG **3750** heel-ward and directing the shaft axis **3791** toe-ward from the CG **3750** may provide a number of performance benefits, including enhanced face rotation dynamics, improved stability on off-center strikes, and increased feel and control. For example, a toe-ward shaft axis **3791** may produce a lever arm between the CG **3750** and the shaft **3790**, potentially resulting in greater responsiveness to a golfer's stroke path and allowing the face portion **3645** to follow the stroke path more intuitively. A toe-ward shaft axis **3791** may help balance the mass distribution of the golf club head **3600**, reducing face twisting on mishits or off-center strikes, thereby enhancing the stability imparted by a heel-ward CG **3750**. The combination of a CG **3750** that is positioned heel-ward and a toe-ward shaft axis **3791** may enhance feedback on mishits and promote a smoother release, potentially improving touch and distance control. Accordingly, the synergistic effects of positioning the CG **3750** heel-ward and directing the shaft axis **3791** toe-ward from the CG **3750** may offer performance benefits that may differ from the performance benefits commonly associated with center-shafted golf clubs and heel-shafted golf clubs that direct a shaft axis through the CG or

heel-ward of the CG. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

In the illustrated example, the hosel portion 3635 may be configured to direct the shaft axis 3791 toward the front-toe quadrant 3921 of FIG. 39. The shaft axis 3791 may intersect the horizontal plane 3760 and the vertical plane 4010 at an intersection point 3799 located toe-ward from the CG 3750. Given the forward lean of the shaft 3790 in the illustrated example, the shaft axis 3791 may extend toward the intersection point 3799 such that a portion of the shaft axis 3791 is located above and forward of the CG 3750 within the front-heel quadrant 3922. Accordingly, the hosel portion 3635 may include an offset to lean the shaft 3790 in a forward direction toward or extending frontward beyond the face portion 3645. By incorporating a forward shaft lean, a golfer's hands may be visually aligned closer toward a golf ball at address, which may encourage a forward press to stabilize the golfer's wrists and promote greater roll consistency. In one example, the shaft axis 3791 and the vertical plane 4010 may define an offset angle 4020 in an x-z plane that is greater than 0 degree and less than or equal to 8 degrees. In another example, the offset angle 4020 may be greater than or equal to 0.5 degree and less than or equal to 3 degrees. In yet another example, the hosel portion 3635 may be configured such that the shaft axis 3791 lies completely on the vertical plane 4010 and extends directly above the CG 3750 within the front-heel quadrant 3922. In yet another example, the hosel portion 3635 may be configured such that the shaft axis 3791 extends toward the intersection point 3799 and a portion of the shaft axis 3791 is located above and rearward of the CG 3750 within the front-heel quadrant 3922. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

In the example of FIGS. 36-41, the intersection point 3799 may have an x-axis value that is opposite and equal or substantially equal in magnitude to the x-axis value of the CG 3750. As defined herein, the term substantially shall mean that a given value is within 15% of a stated value, unless otherwise noted. Accordingly, the CG 3750 and the intersection point 3799 may be equidistant or substantially equidistant from the center longitudinal plane 3720. In the illustrated example, the intersection point 3799 and the CG 3750 may have opposite and equal or substantially equal x-axis values, equal or substantially equal y-axis values, and equal or substantially equal z-axis values. Thus, the CG 3750 and the intersection point 3799 may mirror or substantially mirror one another about the center longitudinal plane 3720. In another example, the intersection point 3799 and the CG 3750 may have x-values, y-values, and/or z-values that vary. Thus, the CG 3750 and the intersection point 3799 may be located at similar or different relative distances from the center longitudinal plane 3720. Accordingly, in other examples, the CG 3750 may be located closer to the center longitudinal plane 3720 and the intersection point 3799 may be located farther away from the center longitudinal plane 3720 or vice versa. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

With respect to any of the examples provided herein, the intersection point 3799 and the CG 3750 may be separated by an offset distance 4030 in a heel-to-toe direction along an axis defined by the intersection between the horizontal plane 3760 and the vertical plane 4010 that is greater than or equal to 0.140 inches (3.556 mm) and less than or equal to 0.260 inches (6.604 mm). In another example, the offset distance 4030 separating the intersection point and the CG may be greater than or equal to 0.160 inches (4.064 mm) and less

than or equal to 0.240 inches (6.096 mm). In yet another example, the offset distance 4030 separating the intersection point and the CG may be greater than or equal to 0.180 inches (4.572 mm) and less than or equal to 0.220 inches (5.588 mm). By locating the intersection point 3799 toe-ward and in near proximity to the CG 3750, a ratio between a moment of inertia (MOI) about a vertical axis of the CG and an MOI about the shaft axis 3791 may be 1 (one) or approximately 1 (one). In one example, the ratio between the MOI about the vertical axis of the CG and the MOI about the shaft axis 3791 may be greater than or equal to 0.80 and less than or equal to 1.20. The ratio between the MOI of the vertical axis of the CG 3750 and the MOI about the shaft axis 3791 may provide an indication of how resistant the golf club head 3600 is to torque, where ratios closer to 1 (one) may signify higher levels of torque resistance and ratios farther away from 1 (one) may signify lower degrees of torque resistance. The aforementioned offset distances 4030 may represent a distance range between the intersection point 3799 and the CG 3750 that may optimize the synergistic effects of positioning the CG 3750 heel-ward and directing the shaft axis 3791 toe-ward from the CG 3750 as described herein. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

FIG. 42 depicts one manner by which any of the golf club heads described herein may be manufactured. In the example of FIG. 42, a process 4200 may begin with providing a golf club head (e.g., golf club head 3600) having a body portion (e.g., body portion 3610) and a hosel portion (e.g., hosel portion 3635) (block 4210). In one example, the hosel portion may be coupled to a hosel receptacle portion (e.g., hosel receptacle portion 3655) provided at a heel portion (e.g., heel portion 3630) of the body portion. Additionally, the hosel portion may have one or more bend portions (e.g., first bend portion 3782 and second bend portion 3784) to impart a forward lean to the shaft. The process 4200 may also include locating a center of gravity (e.g., CG 3750) of the golf club head in a heel-ward direction (block 4220). As described herein, a heel-ward CG may offer enhanced stability and feedback along with improved alignment for many golfers. The process 4200 may further include configuring the hosel portion to direct a shaft axis (e.g., shaft axis 3791) of a shaft (e.g., shaft 3790) in a toe-ward direction from the center of gravity (block 4230). As described herein, the synergistic effects of positioning the center of gravity heel-ward and directing the shaft axis toe-ward from the center of gravity may provide a number of performance benefits, including enhanced face rotation dynamics, improved stability on off-center strikes, and increased feel and control. These performance benefits may impart versatility to the golf club head, making the golf club head a suitable choice for many golfers, particularly those who struggle with squaring the face at impact. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

The filler materials described herein may include, but are not limited to, one or more polymer materials. In one example, the filler material may include an elastic polymer or an elastomer material (e.g., a viscoelastic urethane polymer material such as Sorbothane® material manufactured by Sorbothane, Inc., Kent, Ohio), a thermoplastic elastomer material (TPE), a thermoplastic polyurethane material (TPU), other polymer material(s), bonding material(s) (e.g., adhesive), and/or other suitable types of materials that may absorb shock, isolate vibration, and/or dampen noise. In another example, the filler material may be one or more thermoset polymers having bonding properties (e.g., one or

more adhesive or epoxy materials). The filler material may also absorb shock, isolate vibration, and/or dampen noise when the golf club head **100** strikes a golf ball. Further, the filler material may be an epoxy material that may be flexible or slightly flexible when cured. In another example, the filler material may include any of the 3M™ Scotch-Weld™ DP100 family of epoxy adhesives (e.g., 3M™ Scotch-Weld™ Epoxy Adhesives DP100, DP100 Plus, DP100NS and DP100FR), which are manufactured by 3M corporation of St. Paul, Minnesota. In another example, the filler material may include 3M™ Scotch-Weld™ Epoxy Adhesive DP100 Plus Clear. In another example, the filler material may include low-viscosity, organic, solvent-based solutions and/or dispersions of polymers and other reactive chemicals such as MEGUM™, ROBOND™, and/or THIXON™ materials manufactured by the Dow Chemical Company, Auburn Hills, Michigan. In another example, the filler material may be LOCTITE® materials manufactured by Henkel Corporation, Rocky Hill, Connecticut. In another example, the filler material may be a polymer material such as an ethylene copolymer material that may absorb shock, isolate vibration, and/or dampen noise when a golf club head strikes a golf ball via the face portion. In another example, the filler material may be a high density ethylene copolymer ionomer, a fatty acid modified ethylene copolymer ionomer, a highly amorphous ethylene copolymer ionomer, an ionomer of ethylene acid acrylate terpolymer, an ethylene copolymer comprising a magnesium ionomer, an injection moldable ethylene copolymer that may be used in conventional injection molding equipment to create various shapes, an ethylene copolymer that can be used in conventional extrusion equipment to create various shapes, an ethylene copolymer having high compression and low resilience similar to thermoset polybutadiene rubbers, and/or a blend of highly neutralized polymer compositions, highly neutralized acid polymers or highly neutralized acid polymer compositions, and fillers. For example, the ethylene copolymer may include any of the ethylene copolymers associated with Dupont™ High-Performance Resin (HPF) family of materials (e.g., DuPont™ HPF AD1172, Dupont™ HPF AD1035, DuPont® HPF 1000 and Dupont™ HPF 2000), which are manufactured by E.I. du Pont de Nemours and Company of Wilmington, Delaware. The DuPont™ HPF family of ethylene copolymers are injection moldable and may be used with conventional injection molding equipment and molds, provide low compression, and provide high resilience, i.e., relatively high coefficient of restitution (COR). In another example, the filler material may be formed from one or more metals or metal alloys, such as aluminum, copper, zinc, and/or titanium. A filler material not specifically described in detail herein may include one or more similar or different types of materials described herein. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

Any of the filler materials described herein may be subjected to different processes during manufacturing of any of the golf club heads described herein. Such processes may include one or more filler materials being heated and/or cooled by conduction, convection, and/or radiation during one or more injection molding processes or post injection molding curing processes. For example, all of the heating and cooling processes may be performed by using heating or cooling systems that employ conveyor belts that move a golf club head described herein through a heating or cooling environment for a period of time as described herein. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

While each of the above examples may describe a certain type of golf club head, the apparatus, methods, and articles of manufacture described herein may be applicable to other types of golf club heads (e.g., a driver-type golf club head, a fairway wood-type golf club head, a hybrid-type golf club head, an iron-type golf club head, a putter-type golf club head, etc.).

Procedures defined by golf standard organizations and/or governing bodies such as the United States Golf Association (USGA) and/or the Royal and Ancient Golf Club of St. Andrews (R & A) may be used for measuring the club head volume of any of the golf club heads described herein. For example, a club head volume may be determined by using the weighted water displacement method (i.e., Archimedes Principle). Although the figures may depict particular types of club heads (e.g., a driver-type club head or iron-type golf club head), the apparatus, methods, and articles of manufacture described herein may be applicable to other types of club head (e.g., a fairway wood-type club head, a hybrid-type club head, a putter-type club head, etc.). Accordingly, any golf club head as described herein may have a volume that is within a volume range corresponding to certain type of golf club head as defined by golf governing bodies. A driver-type golf club head may have a club head volume of greater than or equal to 300 cubic centimeters (cm³ or cc). In another example, a driver-type golf club head may have a club head volume of 460 cc. A fairway wood golf club head may have a club head volume of between 100 cc and 300 cc. In one example, a fairway wood golf club head may have a club head volume of 180 cc. An iron-type golf club head may have a club head volume of between 25 cc and 100 cc. In one example, an iron-type golf club head may have a volume of 50 cc. Any of the golf clubs described herein may have the physical characteristics of a certain type of golf club (i.e., driver, fairway wood, iron, etc.), but have a volume that may fall outside of the above-described ranges. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

Any of the golf club heads and/or golf clubs described herein may include one or more sensors (e.g., accelerometers, strain gauges, etc.) for sensing linear motion (e.g., acceleration) and/or forces in all three axes of motion and/or rotational motion (e.g., angular acceleration) and rotational forces about all three axes of motion. In one example, the one or more sensors may be internal sensors that may be located inside the golf club head, the hosel, the shaft, and/or the grip. In another example, the one or more sensors may be external sensors that may be located on the grip, on the shaft, on the hosel, and/or on the golf club head. In yet another example, the one or more sensors may be external sensors that may be attached by an individual to the grip, to the shaft, to the hosel, and/or to the golf club head. In one example, data collected from the sensors may be used to determine any one or more design parameters for any of the golf club heads and/or golf clubs described herein to provide certain performance or optimum performance characteristics. In another example, data from the sensors may be collected during play to assess the performance of an individual. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

Any of the apparatus, methods, or articles of manufacture described herein may include one or more visual identifiers such as alphanumeric characters, colors, images, symbols, logos, and/or geometric shapes. For example, one or more visual identifiers may be manufactured with one or more portions of a golf club such as the golf club head (e.g., casted or molded with the golf club head), painted on the golf club

head, etched on the golf club (e.g., laser etching), embossed on the golf club head, machined onto the golf club head, attached as a separate badge or a sticker on the golf club head (e.g., adhesive, welding, brazing, mechanical lock(s), any combination thereof, etc.), or any combination thereof. The visual identifier may be made from the same material as the golf club head or a different material than the golf club head (e.g., a plastic badge attached to the golf club head with an adhesive). Further, the visual identifier may be associated with manufacturing and/or brand information of the golf club head, the type of golf club head, one or more physical characteristics of the golf club head, or any combination thereof. In particular, a visual identifier may include a brand identifier associated with a manufacturer of the golf club (e.g., trademark, trade name, logo, etc.) or other information regarding the manufacturer. In addition, or alternatively, the visual identifier may include a location (e.g., country of origin), a date of manufacture of the golf club or golf club head, or both.

The visual identifier may include a serial number of the golf club or golf club head, which may be used to check the authenticity to determine whether or not the golf club or golf club head is a counterfeit product. The serial number may also include other information about the golf club that may be encoded with alphanumeric characters (e.g., country of origin, date of manufacture of the golf club, or both). In another example, the visual identifier may include the category or type of the golf club head (e.g., 5-iron, 7-iron, pitching wedge, etc.). In yet another example, the visual identifier may indicate one or more physical characteristics of the golf club head, such as one or more materials of manufacture (e.g., visual identifier of "Titanium" indicating the use of titanium in the golf club head), loft angle, face portion characteristics, mass portion characteristics (e.g., visual identifier of "Tungsten" indicating the use of tungsten mass portions in the golf club head), interior cavity and filler material characteristics (e.g., one or more abbreviations, phrases, or words indicating that the interior cavity is filled with a polymer material), any other information that may visually indicate any physical or play characteristic of the golf club head, or any combination thereof. Further, one or more visual identifiers may provide an ornamental design or contribute to the appearance of the golf club, or the golf club head.

Any of the golf club heads described herein may be manufactured by casting from metal such as steel. However, other techniques for manufacturing a golf club head as described herein may be used such as 3D printing or molding a golf club head from metal or non-metal materials such as ceramics.

All methods described herein may be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. Although a particular order of actions may be described herein with respect to one or more processes, these actions may be performed in other temporal sequences. Further, two or more actions in any of the processes described herein may be performed sequentially, concurrently, or simultaneously.

The terms "and" and "or" may have both conjunctive and disjunctive meanings. The terms "a" and "an" are defined as one or more unless this disclosure indicates otherwise. The term "coupled," and any variation thereof, refers to directly or indirectly connecting two or more elements chemically, mechanically, and/or otherwise. The phrase "removably connected" is defined such that two elements that are "removably connected" may be separated from each other without breaking or destroying the utility of either element.

The term "substantially" when used to describe a characteristic, parameter, property, or value of an element may represent deviations or variations that do not diminish the characteristic, parameter, property, or value that the element may be intended to provide. Deviations or variations in a characteristic, parameter, property, or value of an element may be based on, for example, tolerances, measurement errors, measurement accuracy limitations and other factors. The term "proximate" is synonymous with terms such as "adjacent," "close," "immediate," "nearby," "neighboring," etc., and such terms may be used interchangeably as appearing in this disclosure.

Recitation of ranges of values herein is merely intended to serve as a shorthand method of referring individually to each separate value falling within the range. Unless otherwise indicated herein, each individual value is incorporated into the specification as if it were individually recited herein. A numerical range defined using the word "between" includes numerical values at both end points of the numerical range. A spatial range defined using the word "between" includes any point within the spatial range and the boundaries of the spatial range. A location expressed relative to two spaced apart or overlapping elements using the word "between" includes (i) any space between the elements, (ii) a portion of each element, and/or (iii) the boundaries of each element.

The use of any and all examples, or exemplary language (e.g., "such as") provided herein is intended merely for clarification and does not pose a limitation on the scope of the present disclosure. No language in the specification should be construed as indicating any non-claimed element essential to the practice of any embodiments discussed herein.

Groupings of alternative elements or embodiments disclosed herein are not to be construed as limitations. Each group member may be referred to and claimed individually or in any combination with other members of the group or other elements disclosed herein. One or more members of a group may be included in, or deleted from, a group for reasons of convenience and/or patentability. When any such inclusion or deletion occurs, the specification is deemed to contain the group as modified thus fulfilling the written description of all Markush groups used in the appended claims.

While different features or aspects of an embodiment may be described with respect to one or more features, a singular feature may comprise multiple elements, and multiple features may be combined into one element without departing from the scope of the present disclosure. Further, although methods may be disclosed as comprising one or more operations, a single operation may comprise multiple steps, and multiple operations may be combined into one step without departing from the scope of the present disclosure.

The apparatus, methods, and articles of manufacture described herein may be implemented in a variety of embodiments, and the foregoing description of some of these embodiments does not necessarily represent a complete description of all possible embodiments. Instead, the description of the drawings, and the drawings themselves, disclose at least one embodiment, and may disclose alternative embodiments.

As the rules of golf may change from time to time (e.g., new regulations may be adopted or old rules may be eliminated or modified by golf standard organizations and/or governing bodies such as the USGA, the R & A, etc.), golf equipment related to the apparatus, methods, and articles of manufacture described herein may be conforming or non-conforming to the rules of golf at any particular time.

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Accordingly, golf equipment related to the apparatus, methods, and articles of manufacture described herein may be advertised, offered for sale, and/or sold as conforming or non-conforming golf equipment. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

Further, while the above examples may be described with respect to golf clubs, the apparatus, methods, and articles of manufacture described herein may be applicable to other suitable types of sports equipment such as a fishing pole, a hockey stick, a ski pole, a tennis racket, etc.

Although certain example apparatus, methods, and articles of manufacture have been described herein, the scope of coverage of this disclosure is not limited thereto. On the contrary, this disclosure covers all apparatus, methods, and articles of manufacture fairly falling within the scope of the appended claims either literally or under the doctrine of equivalents.

What is claimed is:

1. A golf club head comprising:

a body portion comprising a toe portion, a heel portion, a front portion, a rear portion, a top portion, a sole portion, and an alignment aid at the top portion, the body portion having a center of gravity located on a horizontal plane that is parallel to a ground plane; and a hosel portion coupled to the body portion at or proximate to the heel portion, the hosel portion comprising a first bend portion, a neck portion, a second bend portion, and a tip portion, the tip portion configured to attach to a shaft comprising a shaft axis that is coaxial with the tip portion,

wherein the center of gravity is located heel-ward from a center longitudinal plane bisecting the body portion to define the toe portion and the heel portion,

wherein the hosel portion is configured to orient the shaft such that the shaft axis intersects the horizontal plane at an intersection point located toe-ward from the center longitudinal plane,

wherein the center of gravity and the intersection point are equidistant or substantially equidistant from the center longitudinal plane,

wherein a visibility angle is formed between the center longitudinal plane and a visibility plane bordering a toe-most extent of the second bend portion and intersecting the center longitudinal plane at the alignment aid,

wherein the visibility angle represents a range of sight-lines that enable an individual to fully see the alignment aid, and

wherein the visibility angle is greater than or equal to 5 degrees and less than or equal to 20 degrees.

2. A golf club head as defined in claim 1, wherein the center of gravity lies within a center of gravity zone bounded by the center longitudinal plane and a boundary plane parallel to the center longitudinal plane and defining a toe-side boundary of the hosel portion.

3. A golf club head as defined in claim 1, wherein the center of gravity and the intersection point are separated by an offset distance greater than or equal to 0.140 inches (3.556 mm) and less than or equal to 0.260 inches (6.604 mm).

4. A golf club head as defined in claim 1, wherein the center of gravity and the intersection point are separated by an offset distance greater than or equal to 0.160 inches (4.064 mm) and less than or equal to 0.240 inches (6.096 mm).

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5. A golf club head as defined in claim 1, wherein the center of gravity and the intersection point are separated by an offset distance greater than or equal to 0.180 inches (4.572 mm) and less than or equal to 0.220 inches (5.588 mm).

6. A golf club head as defined in claim 1, wherein the center of gravity is located closer to the top portion than to the sole portion, and wherein the center of gravity is located closer to the front portion than to the rear portion.

7. A golf club head as defined in claim 1, wherein the first bend portion is configured to direct the neck portion in a toe-ward direction and a rearward direction, and wherein the second bend portion is configured to direct the tip portion in a heel-ward direction and a frontward direction.

8. A golf club head as defined in claim 1, configured as a mallet type putter head.

9. A golf club head comprising:

a body portion comprising a toe portion having a toe portion edge, a heel portion having a heel portion edge, a front portion having a face portion, a rear portion having a rear portion edge, a top portion having a top portion edge, a sole portion having a sole portion edge, and an alignment aid at the top portion, the body portion having a center of gravity located on a horizontal plane that is parallel to a ground plane; and

a hosel portion coupled to the body portion and located heel-ward of a center longitudinal plane bisecting the body portion to define the toe portion and the heel portion, the hosel portion comprising a first bend portion, a neck portion extending from the first bend portion, a second bend portion extending from the neck portion, and a tip portion extending from the second bend portion, the tip portion configured to attach to a shaft comprising a shaft axis that is coaxial with the tip portion,

wherein the first bend portion is configured to direct the neck portion in a toe-ward direction toward the toe portion edge and a rearward direction toward the rear portion edge,

wherein the second bend portion is configured to direct the tip portion in a heel-ward direction toward the heel portion edge and a frontward direction toward the face portion,

wherein the shaft axis intersects the horizontal plane at an intersection point located toe-ward from the center of gravity,

wherein a visibility angle is formed between the center longitudinal plane and a visibility plane bordering a toe-most extent of the second bend portion and intersecting the center longitudinal plane at the alignment aid,

wherein the visibility angle represents a range of sight-lines that enable an individual to fully see the alignment aid, and

wherein the visibility angle is greater than or equal to 5 degrees and less than or equal to 20 degrees.

10. A golf club head as defined in claim 9, wherein a portion of the shaft axis extends above and forward of the center of gravity.

11. A golf club head as defined in claim 9, wherein the center of gravity is located closer to the heel portion edge than to the toe portion edge.

12. A golf club head as defined in claim 9, wherein the center of gravity and the intersection point are separated by an offset distance greater than or equal to 0.180 inches (4.572 mm) and less than or equal to 0.220 inches (5.588 mm).

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13. A golf club head as defined in claim 9, wherein the hosel portion further comprises an insert portion configured to be received inside a hosel bore formed at the heel portion.

14. A golf club head as defined in claim 9, configured as a mallet type putter head.

15. A golf club head comprising:

a body portion comprising a toe portion, a heel portion, a front portion, a rear portion, a top portion, a sole portion, and an alignment aid at the top portion, the body portion having a center of gravity located on a horizontal plane that is parallel to a ground plane;

a hosel portion coupled to the body portion, the hosel portion comprising a first bend portion, a neck portion, a second bend portion, and a tip portion, the tip portion configured to attach to a shaft comprising a shaft axis, wherein the shaft axis intersects the horizontal plane at an intersection point,

wherein a portion of the shaft axis extends above and forward of the center of gravity,

wherein the center of gravity and the intersection point are equidistant or substantially equidistant from a center longitudinal plane bisecting the body portion to define the toe portion and the heel portion,

wherein a visibility angle is formed between the center longitudinal plane and a visibility plane bordering a

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toe-most extent of the second bend portion and intersecting the center longitudinal plane at the alignment aid,

wherein the visibility angle represents a range of sight-lines that enable an individual to fully see the alignment aid, and

wherein the visibility angle is greater than or equal to 5 degrees and less than or equal to 20 degrees.

16. A golf club head as defined in claim 15, wherein the first bend portion is configured to direct the neck portion in a toe-ward direction and a rearward direction, and wherein the second bend portion is configured to direct the tip portion in a heel-ward direction and a frontward direction.

17. A golf club head as defined in claim 15, wherein the center of gravity and the intersection point are separated by an offset distance greater than or equal to 0.180 inches (4.572 mm) and less than or equal to 0.220 inches (5.588 mm).

18. A golf club head as defined in claim 15, wherein the intersection point is located to mirror or substantially mirror the center of gravity about the center longitudinal plane.

19. A golf club head as defined in claim 15, wherein the intersection point is located toe-ward from the center of gravity.

20. A golf club head as defined in claim 15, configured as a mallet type putter head.

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